Behind the Lens:
Making Home CCTV Data Practices Visible

Checkpoint 3: Exhibit

Project exhibit and documentation

DECO7380

Team: ANN

Binyan Zhang 48878162 Xinyue Chang 49372528

Yining Zhu 49053881 Yuxuan Yang 48950255

27 October 2026

AIRIS Privacy Guidance and Shared Access for Home CCTV

#UX/UI #HomeCCTV #UsablePrivacy #UXDesign #AccessControl #WebApplication

Project Brief

This project aims to help household CCTV owners understand how footage is stored, transmitted and accessed, and make informed choices about shared access. It connects AIRIS, an independent privacy guidance platform, with a conceptual eufy interface redesign. Users ask a specific privacy question, inspect a concise explanation and its sources, and follow a labelled route to relevant settings. The eufy concept demonstrates shared viewing permissions and Access History. Further development should investigate whether this connected experience adds value beyond in-app guidance. Implementing changes in the official eufy app requires manufacturer cooperation; the current prototype uses simulated actions and records.

Problem Space

Home cameras are easy to install and view, but their data practices are hard to understand. Owners may not know whether footage passes through the cloud, under what conditions the manufacturer can access it, or what shared family members can see or change. Our interviews found that privacy information was difficult to locate and long to read, and that adding a member did not mean understanding their permissions. Earlier label testing showed that summary ratings were easily misread without visible meaning and evidence. Because cameras also record people without accounts, these gaps affect more than the device owner. Existing applications provide controls, but rarely connect explanations with the actions users need to take.

Success Criteria

Owners can independently find privacy explanations and settings, identify a shared member’s effective access, distinguish viewing records from recording history, and explain what permission changes affect. Evaluation should test task completion and understanding, including recognition of simulated actions, rather than preference alone.

This Application MUST Have

- Explain where footage is stored and who can access it, with sources
- Link each privacy question to the right setting
- Show what each shared member can see
- Explain the consequences before a permission changes
- Show who viewed each camera and when
- Make clear which actions change a real device

This Application COULD Have

- Step-by-step visual tutorials for verified settings.
- Additional CCTV models with reviewed product documentation.
- Separate login and settings-change records.
- Validated masking controls and reliable event integration, subject to manufacturer support.

Expected Skills

- Frontend web development (HTML, CSS, JavaScript)
- UX/UI design and usability evaluation
- Usable privacy and security research
- API or manufacturer integration

Contents

1 Project Overview *4*

1.1 The CCTV Privacy Problem and Project Purpose 4

1.2 Target Users and the AIRIS and eufy Experience 4

1.3 Main Changes Since the Interim Critique 4

2 Research Evidence and Stakeholder Needs *5*

2.1 Research Findings Informing Prototype Iteration 5

2.2 Needs of Owners, Shared Users and People Being Recorded 6

2.3 Literature and Requirements Informing the Updated Design 7

3 Prototype Development and Iteration *8*

3.1 Design Baseline at the Interim Checkpoint 8

3.2 Additional Interviews and Revised Design Priorities 9

3.3 Key Changes from the Interim Proposal to the Web Prototype 10

3.4 Current Web Prototype and the Connected User Journey 12

3.5 What Changed Across the Iterations and Why 14

4 Ethical Considerations *15*

4.1 Household Authority and Bystander Privacy 15

4.2 Trustworthy Guidance and Informed Choices 15

4.3 Privacy Risks of Access Records and Viewer Verification 16

5 Evaluation and Build Handover *16*

5.1 Completed Evaluation and Its Limitations 16

5.2 Evaluation Tasks and Success Criteria 17

5.3 Functional and Non-Functional Requirements 17

5.4 Project Risks and Prioritised Questions for the Build Team 19

6 Team Process and Critical Reflection *20*

6.1 Project Milestones and Changes to the Original Plan 20

6.2 How Feedback Changed Our Decisions 20

6.3 Collaboration Strengths, Challenges and Lessons 21

6.4 Final Design Position 21

References *22*

Appendices *24*

Appendix A - Interview Materials and Evidence Index 24

Appendix B - Round 1 Label Comparison 30
Appendix C - Round 2 Cross Product Comparison 34
Appendix D - Affinity Mapping 37
Appendix E - Evidence to Design and Evaluation Index 37
Appendix F - AIRIS Interface Screens 38
Appendix G - Current eufy Prototype Screens 41
Appendix H – Product Privacy Labels 48
Appendix I - AI Acknowledgement Cover Sheet 49
Appendix J - Product Information Sources 51
Appendix K – Prototype Repository 52
Appendix L – Exhibition Poster 53

1 Project Overview

1.1 The CCTV Privacy Problem and Project Purpose

Home CCTV systems allow users to remotely monitor their homes, keep an eye on their pets, and ensure the safety of their family members. However, using a home CCTV system does not necessarily mean users know or understand how the footage is transmitted, stored, and accessed. Users may know where to view the live feed, but they may not be clear on whether the video is transmitted via the cloud, under what conditions the manufacturer can access it, or where to adjust the relevant settings. When users have questions, they may not know where to find answers or what steps to take. Since cameras may also record the activities of housemates, visitors, and others, these uncertainties affect more than just the device owner.

We aim to help users clarify these issues and translate that understanding into practical choices. For example, users need to know not only that they can share camera feeds with family members, but also what those members can see, how to restrict their access, and what happens after permissions are modified. Therefore, this project centres on “understanding, finding, choosing and reviewing”: helping users understand how the camera processes data, locate relevant privacy settings, independently decide what others can see, and review access activity when needed. We want privacy management to be integrated into everyday use, rather than appearing only during initial setup or being buried in hard-to-understand instructions.

1.2 Target Users and the AIRIS and eufy Experience

Our primary target users are household CCTV owners who manage shared access but are unfamiliar with privacy settings. We aim to help them understand data handling, inspect shared permissions and explore access records. Shared-access members, co-residents and visitors are also affected by owners’ decisions.

AIRIS provides question-based privacy explanations, data-flow views and links to relevant settings. Step-by-step tutorials are planned. The conceptual eufy redesign demonstrates privacy statements, shared viewing permissions and Access History. Together they connect asking a question, understanding an explanation, finding settings, making a choice and reviewing access. AIRIS can be developed independently of manufacturer changes, but its value beyond in-app guidance remains untested. Our website uses sample identities and simulated interactions; it cannot change real eufy settings. Deployment would require manufacturer collaboration. Restricting future access would also not remove copies already downloaded.

We chose eufy because it appeared in our interim label testing and publishes documentation that explanations can cite. Historical reporting on cloud thumbnails and web-stream encryption illustrates why privacy claims need conditions (Lovejoy, 2022, 2023). It does not establish a vulnerability in current models. Appendix J distinguishes current manufacturer statements from historical reports.

1.3 Main Changes Since the Interim Critique

Our focus this time was on connecting previously separate design concepts. The project evolved from paper privacy labels into a web prototype demonstrating key stages of a proposed user journey. Alongside information presentation, we explored how users could locate settings and manage access after understanding an explanation.

The AIRIS section now includes problem-oriented explanations, data-flow visualisations, and relevant action entry points, along with a planned step-by-step navigation tutorial. The eufy section features improvements centered on privacy statements, shared viewing permissions, and access logs, enabling users to view information and make choices in specific contexts.

Six additional interviews extended our understanding of app use, settings and sharing. Interim label testing (Rounds 1 and 2) highlighted unclear access conditions and difficulties interpreting ratings. These findings

informed the updated explanations and permission views; the connected website and activity-history interactions still require evaluation.

2 Research Evidence and Stakeholder Needs

2.1 Research Findings Informing Prototype Iteration

We combined observations from the interim label testing (Rounds 1 and 2; Section 3.1) with six additional interviews (P13–P18) to identify five findings that informed the transition from paper prototypes to the web prototype. The additional interviews brought the total to 18 interview records. P13 described store use; the other five participants discussed household use. As the interview questions differed between rounds, responses were analysed thematically rather than treated as comparable survey data. Testing observations, early interview evidence and follow-up interview quotations are distinguished below.

Interview sources and the links between evidence, findings, design requirements and evaluation are indexed in Appendices A and E.

2.1.1 Finding 1 Privacy Explanations Need Specific Facts and Clear Conditions

The interim label tests highlighted uncertainty around conditional privacy statements. Test participant R1-P02 asked: *"Ah, what do you mean by 'for a specific purpose'?"* The follow-up interviews identified a related information need. P15 stated*: "I want to know if the camera company can see my recordings. But the privacy information wasn't very easy to find."* [Source: Appendix A3]

These findings suggest that broad assurances are insufficient. Explanations should distinguish storage locations, access roles and relevant conditions. This informed the move from compact paper labels towards browsable privacy information and data-flow explanations in the web prototype. Product-specific claims must remain linked to verifiable sources.

2.1.2 Finding 2 Visual Summaries Need Clear Meanings and Supporting Evidence

The interim label records identified uncertainty about score meanings and direction. In the available R2-P06 transcript, the participant distinguished lower Exposure from higher User Control, but reported initial hesitation because high scores usually seem preferable. This is an observation of one available record, not a count of misinterpretations across all six sessions.[Source: Appendix A3]

Our interpretation is that visual summaries should introduce information without replacing the underlying facts. The updated design therefore connects privacy summaries with explanations and supporting sources. Any retained scores must clearly communicate their meaning and basis; illustrative prototype values should not appear to be independently verified product assessments.

2.1.3 Finding 3 Guidance Should Be Easy to Find and Offer Optional Detail

P16 described difficulty finding an answer: *"I looked in the settings, but I couldn't find a clear answer. There was a lot of text, so I didn't read everything."* They also requested flexibility: *"It would also be helpful if I could click to read more details when I want to."* P15 preferred privacy features *"in the same CCTV app instead of using a separate app."*[Source: Appendix A3]

This supports concise, question-based explanations connected to relevant settings. It informed layered content across AIRIS and eufy, consistent with Schaub et al. (2015) privacy notice design space. However, these responses favour accessible guidance rather than directly supporting a separate platform. AIRIS's additional value and the effort of switching interfaces remain evaluation questions.

Emami-Naeini et al. (2020) developed a two-layer IoT privacy and security label with experts and consumers. This provides a precedent for concise summaries with inspectable detail, rather than evidence that our specific layout improves understanding.

2.1.4 Finding 4 Sharing Interfaces Should Make Access Boundaries Visible

P16 was *"not always sure what other people can see or change"*, adding: *"I wish the app explained this more clearly."* P17 requested *"a clear page showing who can see the camera."*[Source: Appendix A3]

Adding a member does not necessarily establish understanding of their permissions. This is consistent with Geeng & Roesner (2019) findings about uneven household knowledge and control. Our design response is a member permission view connecting people with cameras and viewing options. This extends the earlier information-focused prototypes towards interactive permission management, although the categories and consequences still require testing.

2.1.5 Finding 5 Viewing Records Address a Different Need from Recording History

Early interviews identified interest in footage-access information. P3 stated: *"I'd want to know who accessed it, when they accessed it and why."* In the follow-up interviews, P15 distinguished existing recording features from a further need: *"My app sends motion detection alerts, and it shows recording history. These are useful, but sometimes I get too many notifications. I would also like to see login history."*[Source: Appendix A3]

Recording history explains what a camera captured, not who viewed it. This informed the proposed Access History, which demonstrates who viewed which camera, when, and whether the visit involved live video or a recording. Past visits remain separate from current permissions. Login history is a different requirement and is not yet covered. P15 provides direct household support for login records; P13's store-use account is an additional contextual reference. These statements motivate exploration rather than establish the event coverage needed by all households.

Together, these findings informed the shift from presenting privacy information on paper to exploring explanations, permissions and activity records through an interactive website. They explain our design decisions without establishing that the updated prototype has resolved the identified problems. The additional interview affinity diagram is provided in Appendix D.

2.2 Needs of Owners, Shared Users and People Being Recorded

The primary target users are household CCTV owners who manage shared access but are unfamiliar with privacy settings. Their role in managing shared access makes them suitable users for evaluating the proposed permission and activity-history flows. This focus is a design decision informed by the interviews, literature and project scope; it does not imply that all participants belong to this group.

Geeng & Roesner (2019) found that device selection, management and maintenance in smart homes are often carried out by the person who installed the devices, leaving knowledge and control unevenly distributed among household members. We therefore cannot assume that other members understand their own permissions. At the same time, our interviews showed that even users responsible for managing cameras may struggle to locate privacy information or confirm sharing permissions (see Findings 3 and 4), so we also cannot assume that managers fully understand the available privacy settings. Zeng & Roesner (2019) designed and deployed a multi-user smart home prototype with access controls and activity notifications, offering a relevant reference for permission control and activity visibility.

Beyond the primary users, the design must also consider the following stakeholders:

Table 1. Stakeholder needs and considerations.

Stakeholder	Relationship to the project and key considerations
Shared-access members	Need to know which cameras they can access, what restrictions apply, and whether their own viewing activity is recorded.
Co-residents and visitors	May appear in footage without having an account or management rights; their privacy should not be determined solely by the owner's convenience.
Manufacturers and service providers	Provide the actual settings, product documentation and event data, which affect whether AIRIS explanations are accurate and whether the proposed controls can be implemented.

Because the proposed Access History would record shared members' viewing activity, future iterations should make this visibility clear to those members, not only to the owner. The Hide faces and Hide whole person options partly respond to people being recorded by limiting what shared viewers can see (Section 3.3.3). Marky et al. (2021) comparison of residents' and visitors' privacy mental models supports distinguishing between people who use a device and people affected by it. However, the current interviews provide limited direct coverage of non-owners and visitors, so owners' views cannot be treated as a complete account of these groups' needs.

2.3 Literature and Requirements Informing the Updated Design

The findings in Section 2.1 informed five design requirements. Table 2 links each requirement to its evidence and proposed design response.

Table 2. Evidence and requirements informing the updated design. See Appendix E for the evidence index.

Requirement	Supporting evidence	Proposed design response
R1 Explain how footage is stored, transmitted and accessed	Finding 1; early interviews on storage, data flow and external access (Figure 3.1)	AIRIS data-flow explanations; eufy privacy guide, illustrated Agreement and Storage & Backup
R2 Help users locate settings relevant to their questions	Finding 3	AIRIS answers with "Continue in eufy" links; step-by-step tutorials specified but not yet implemented
R3 Make shared members' access boundaries clear	Finding 4	Viewing Permissions with per-camera visibility options; Camera Events requests for individual clips
R4 Provide records of footage access	Finding 5; early interviews (P3); login history requested by P15 only among households	Access History of live and recording visits, separate from current permissions; login records not yet covered
R5 Provide concise summaries with clear meanings, supporting sources and optional detail	Findings 2 and 3	Layered explanations, expandable sources and clear labels for illustrative scores

Zheng et al. (2018) found that convenience and trust shaped smart-home owners' privacy practices. Manandhar et al. (2022) identified gaps in the availability, content and coverage of smart-home privacy policies. These studies support accessible, source-specific explanations while cautioning against treating a policy summary as proof of actual device behaviour.

Feng et al. (2021) distinguish privacy choices delivered through primary and secondary channels. Our design adapts this distinction: AIRIS provides explanations and guidance, while the eufy prototype supports viewing

and changing simulated settings. This approach allows guidance to be developed independently of manufacturer changes, although its value compared with an integrated app remains unproven.

PrivacyHub demonstrates a functional system combining data-flow awareness with device-connectivity controls (Windl et al., 2025). It is a related design precedent, not validation of AIRIS: our prototype neither controls device connectivity nor establishes cross-brand interoperability.

Farke et al. (2021) show that greater comfort with data practices does not necessarily translate into intended action. Evaluation will therefore examine whether users understand data handling, locate permissions and complete relevant tasks, alongside their interface preferences.

3 Prototype Development and Iteration

3.1 Design Baseline at the Interim Checkpoint

At the interim checkpoint, the design was organised around SEE, UNDERSTAND, CONTROL and REVIEW. These stages connected four proposed components: Privacy Label, Privacy Guide, Privacy Settings and Access History. Their purpose was to support both understanding of CCTV data practices and decisions about access.

Selected early records explain why we retained concise information, access conditions and reviewable activity (Figure 3.1). P3, a prospective buyer, wanted to know *“who accessed it, when they accessed it and why”*. P7 also requested an access record, while P9 preferred a simple in-app privacy page. These examples support design exploration, not an estimate of how common each need is.

Low-fidelity prototypes represented these needs through a label containing privacy facts and summary ratings, a simplified data journey, proposed settings and example access records. The label combined overview ratings and concrete facts. This fidelity allowed information grouping and terminology to be explored without implying working camera connections or permission enforcement.

Evaluation concentrated on the label. Round 1 compared three versions containing the same product information: Facts Only, Rating Label and XY Privacy Profile. Six participants (R1-P01–R1-P06) answered questions about access, storage and control, explained their reasoning and ranked the formats; the checkpoint summary reports four first-place preferences for Rating Label, two for Facts Only and none for XY Privacy Profile. Round 2 examined labels for three camera products. Participants interpreted each label before reviewing the scoring guide and supporting evidence. The available R2-P06 record documents initial hesitation over opposite score directions and uncertainty about score calculation. We therefore retained the Rating Label while pairing ratings with factual explanations and inspectable evidence (Findings 1 and 2). Preference did not establish superior comprehension, and the small samples and repeated exposure limit these conclusions. Detailed Round 1 and Round 2 summary records are provided in Appendices B and C.Detailed Round 1 and Round 2 summary records are provided in Appendices B and C.

The resulting baseline was stronger as an information framework than as a demonstrated management journey. Whether users could move between components, select an appropriate control and understand its consequences remained unresolved. These were questions for subsequent development, not outcomes established by the label tests.



Figure 3.1. Selected early interview evidence linked to the interim design. These are illustrative source extracts, not frequency counts. Sources and selection rules are given in Appendix A5.

3.2 Additional Interviews and Revised Design Priorities

The six additional interviews (P13–P18) informed the priorities below. P13 described store use and the other five described household use. Findings 3–5 (Section 2.1) identify information-finding, permission and activity-record needs; Figure 3.2 provides a descriptive summary. These accounts concerned existing CCTV applications, not testing of the final website. Participant-level support is provided in Appendix E2; interview records are in Appendix A4.

Finding 3 led us to prioritise accessible explanations linked to relevant settings. P15 preferred privacy features *"in the same CCTV app instead of using a separate app."* This challenges the separate AIRIS platform and makes its additional value an evaluation question. Clear destination labels are a proposed response, not evidence that switching is effortless.

Finding 4 prioritised visible access boundaries: users should be able to inspect the person, camera and viewing scope. P13's staff-permission needs extend this question into a workplace setting but cannot represent household requirements. The exact permission categories remain design hypotheses.

Finding 5 led us to distinguish recording history, footage-viewing records and account logins. Earlier access-related evidence supports exploring viewing records; P15's additional request concerns login history, which the current demonstration does not provide. Notification burden remains outside scope and does not validate access logging as its solution.

Finding 3 also informed concise explanations with optional detail. P16 stated: *"It would also be helpful if I could click to read more details when I want to."* Such preferences support exploring layered presentation, not claims that pictures or icons have improved comprehension.

These findings retained the four-stage framework while clarifying its practical requirements: help users find information, understand access boundaries, distinguish activity types and interpret explanations before

deciding. Section 3.3 explains how these priorities informed interface changes. The interviews provided reasons for those changes; the effectiveness of the final web interactions requires separate evaluation.

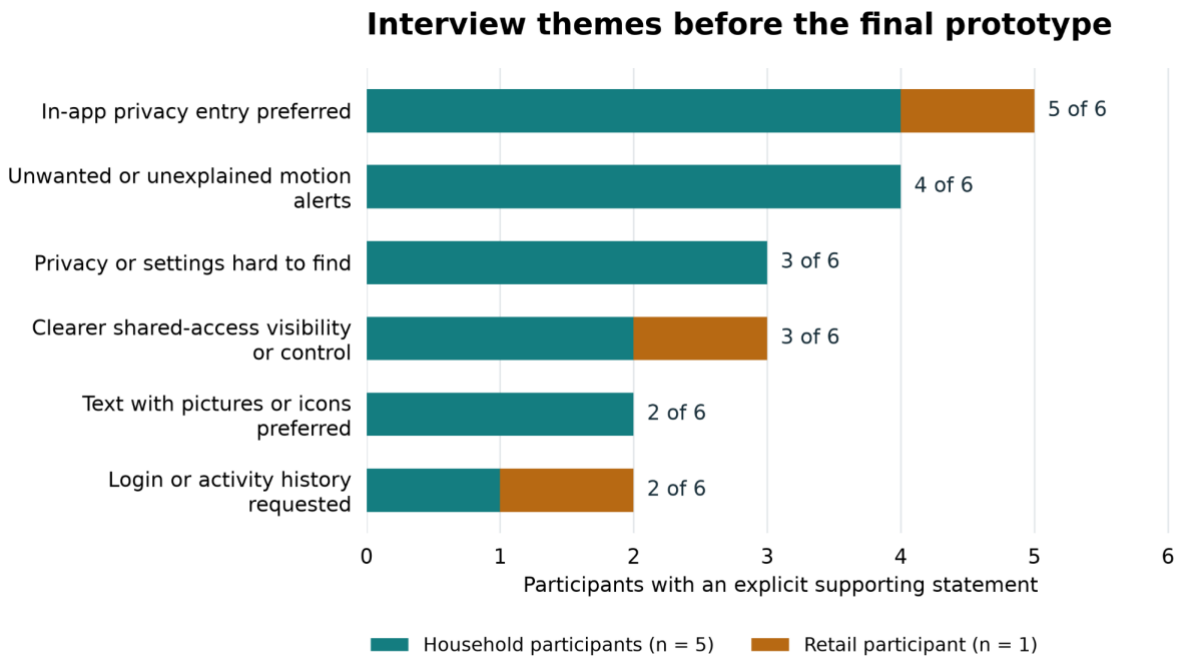


Figure 3.2. Interview themes from the six additional interviews (P13–P18).

Counts show participants with an explicit supporting statement. Questions varied and themes overlap, so counts are not feature-approval votes. These interviews did not test the final prototype.

3.3 Key Changes from the Interim Proposal to the Web Prototype

The interim proposal established a four-stage privacy journey: SEE, UNDERSTAND, CONTROL and REVIEW. However, its labels, explanations and feature sketches did not fully show how users could move from identifying a concern to taking action. Following the interviews discussed in Section 3.2, we retained this framework and strengthened the connections between information, privacy choices and access records.

3.3.1 Connecting Privacy Information to Relevant Actions

The earlier Privacy Label helped users compare product privacy information. However, understanding a product's practices did not necessarily help existing CCTV owners find the relevant settings. P15 described difficulty finding privacy information, while P17 preferred managing privacy within the camera application. These accounts highlighted the need to connect explanations with accessible controls.

AIRIS therefore adds a question-based entry point alongside its product privacy profiles. Users can begin with a question about camera access, read an explanation and follow a link to the relevant Access History or permission page in the eufy prototype. The profiles and supporting evidence remain available, so users do not have to discover every important privacy issue through their own questions.

The supplied AIRIS screens also show a Privacy Profile with separate Exposure and User Control directions, links to data-flow and evidence views, and a "Continue in eufy" action (Appendix F). The profile labels its scores as historical research material rather than safety certification. These elements make the interpretation and next action inspectable; their effectiveness remains untested.

This introduces a trade-off. AIRIS provides space for independent explanations, but switching between websites adds steps and conflicts with some participants' preference for one application. The current design therefore places explanations and entry points in AIRIS, while keeping permission decisions within eufy. Whether users understand this handover still needs evaluation.

3.3.2 Explaining Privacy at Relevant Moments

The earlier Privacy Guide used a simplified data journey to explain the relationship between cameras, storage services and applications. This development considered when users should encounter those explanations. Before users enter the demonstration, the current Agreement introduces collection, storage, access and deletion through short text, animations and illustrations. Expandable details and a full-policy link provide further information. P16 stated, *"I prefer short text with simple pictures or icons,"* while P18 also suggested combining text and images during initial use. These accounts support exploring this format, but do not establish that the current design improves understanding.

Users can also reopen an optional privacy guide through Settings. The initial explanation and later guide therefore serve different moments of use. Reopening the guide does not change permissions or constitute renewed consent. This layered approach is consistent with Schaub et al. (2015), who discuss the importance of notice timing and presentation. However, scrolling, waiting or ticking a confirmation box records an interaction; it does not prove understanding.

3.3.3 Making Access Permissions and Their Scope Visible

The interim report identified a conflict: automatic masking could protect people recorded by a camera but reduce the usefulness of footage after an incident. The current prototype therefore presents masking as a choice within specific access situations, rather than applying it universally.

Viewing Permissions identifies the member and camera concerned, then offers original view, hidden faces, hidden whole person and no access. An illustrated preview shows the intended difference. This responds to P16's uncertainty: *"I'm not always sure what other people can see or change."* Camera Events similarly presents individual recording requests with the requester, stated purpose and options to allow or deny access. It distinguishes permission for a particular clip from broader camera access.

The prototype also includes a simulated viewer-verification step. It explores whether a camera could confirm that the person opening a feed matches a permitted viewer, separating identity checks from permission decisions. Participants did not request this feature; it is an exploratory extension of permission boundaries that uses no real biometric data. Its privacy risks are discussed in Section 4.3.

These changes aim to clarify who can see what and under which conditions. However, the interviews support clearer permission boundaries, rather than confirming demand for these exact masking options. The camera owner's decision also cannot represent everyone recorded. The previews demonstrate the proposed interaction; they do not verify masking accuracy or secure enforcement.

3.3.4 Clarifying the Consequences of Storage and Access Decisions

Storage settings and Access History were already part of the interim proposal. Their further development focused on explaining the consequences of decisions and connecting records to follow-up actions.

Storage & Backup presents local storage, cloud storage, automatic backup and retention separately. Local and cloud storage can coexist, rather than being treated as mutually exclusive choices. The interface also explains that stopping future storage or backup does not delete existing recordings. Deletion therefore requires a separate action. The cloud backup request in Camera Events makes the proposed data route visible and provides an explicit choice to allow or decline uploading.

Access History develops the earlier example records into expandable entries. Users can inspect the viewer and camera, distinguish permission at the time of access from current permission, and open member settings to restrict future access. This extends P3's request: *"I'd want to know who accessed it, when they accessed it and why."* It also avoids suggesting that changing permissions now changes an earlier visit.

The current history mainly demonstrates footage-viewing records. It does not yet cover P15's requested login history or the broader account activities discussed by P13. The records and previews are illustrative, and the controls do not operate a real camera or cloud service. These changes translate research findings into concrete interactions, but their effectiveness remains untested. Section 3.4 shows how they connect within the current user journey.

3.4 Current Web Prototype and the Connected User Journey

The current prototype connects understanding a privacy concern with reviewing and changing access. AIRIS explains privacy issues and points users towards relevant controls, while the eufy prototype demonstrates how those controls could work. The main scenario follows a household CCTV owner managing shared viewing. Shared viewers and people recorded by the camera are also affected, so the owner's choices cannot represent everyone's privacy preferences.

The journey begins when the owner asks AIRIS, "Who can access my camera?" The answer introduces example viewers, explains the difference between current permissions and previous visits, and offers further detail when needed. Links labelled "Continue in eufy" name their destinations, including Access History and a viewer's permissions. The owner can therefore see where the next action will take place. Reading an AIRIS answer does not itself grant or remove camera access.

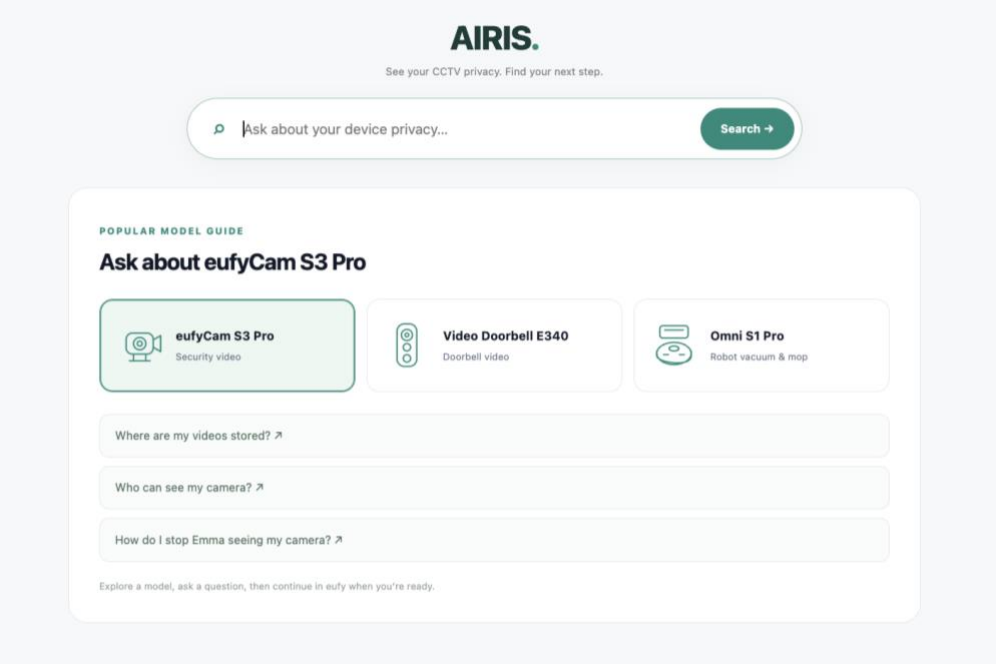


Figure 3.4a. AIRIS model guide offers example questions about storage, access and restricting a shared viewer. The screen is a student research prototype.

The AIRIS model guide offers selectable questions alongside the search field (Figure 3.4a), allowing users to start without composing their own query. The illustrated scenario focuses on home CCTV; the robot-vacuum entry is outside this report's evaluation scope.

The owner then opens Access History within Events, selects a date and expands a viewer's visits (Figure 3.4b). This develops an earlier need expressed by P3: "I'd want to know who accessed it, when they accessed

it and why." Each entry separates the time of access from the time the footage was recorded. It also distinguishes "Allowed at visit" from the viewer's current permission. This matters because changing a permission now should not change how an earlier visit is represented. The prototype mainly shows footage-viewing records; it does not yet provide the account-login history requested by P15.

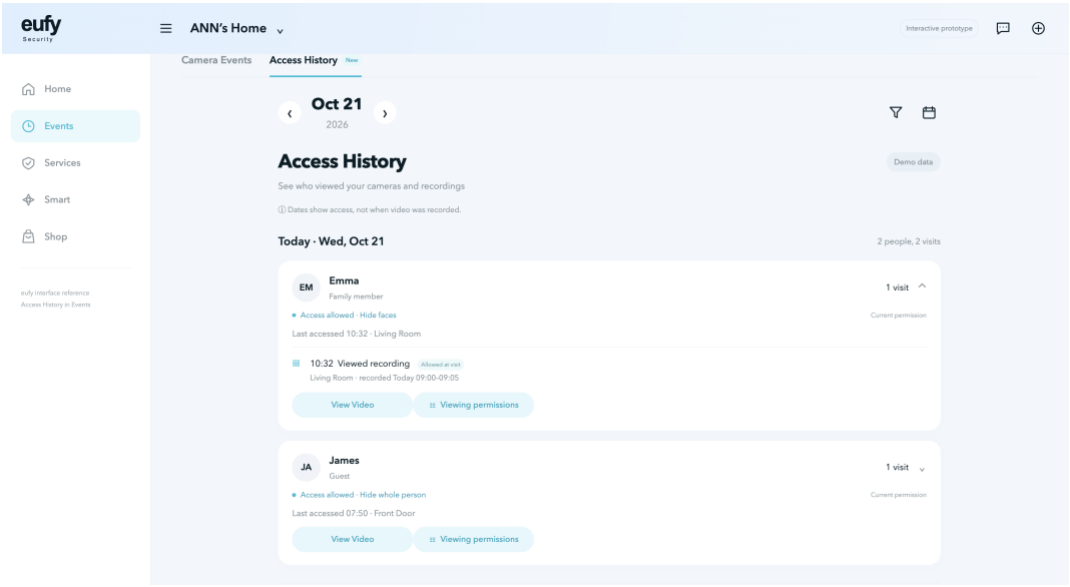


Figure 3.4b. Latest Access History separates current permission from "Allowed at visit" and links visits to Viewing permissions. All people and events are demonstration data.

From the record, the owner follows the Viewing permissions action in Figure 3.4b and chooses Original view, Hide faces, Hide whole person or No access. An illustrated preview shows the intended visibility level. The selected option and current-permission label let the owner check the demonstrated result. These choices address the broader need for clearer shared access, but the interviews did not establish a preference for these exact masking modes. Changes apply to future access; they do not remove earlier visits or recover downloaded copies. Owner approval also does not establish agreement from everyone recorded.

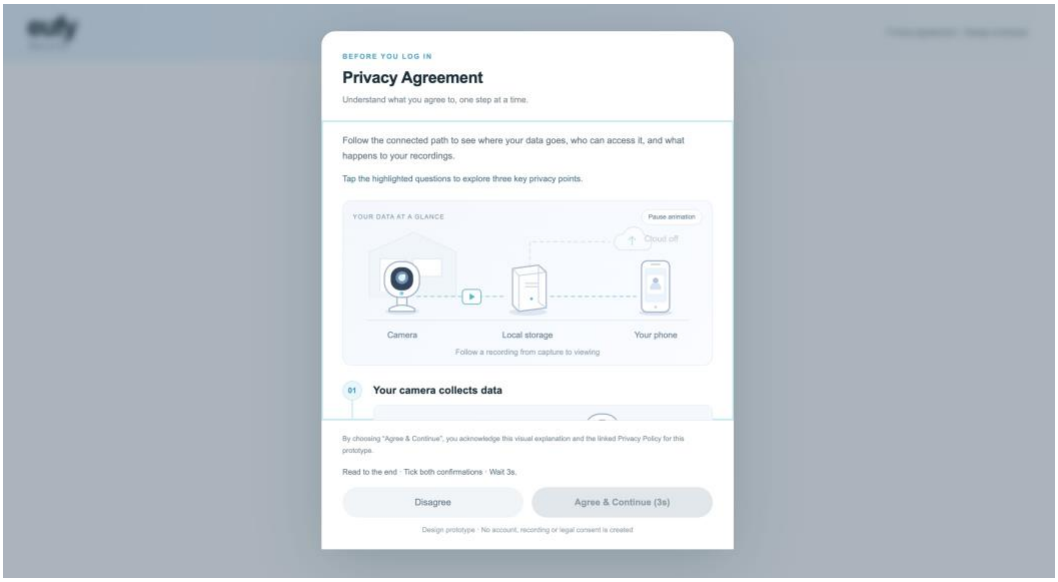


Figure 3.4c. Latest Privacy Agreement introduces collection, storage, access and deletion before demo entry. Disagree and Agree & Continue are separate actions; the screenshot does not establish understanding.

Other interactions follow the same approach of explaining a choice before asking for a decision. Camera Events identifies the requester, clip and purpose, while keeping personal viewing requests separate from

cloud-backup approval. Pending requests are not records of completed viewing. Storage offers separate local and cloud controls and explains that stopping future storage does not delete existing recordings. Before demo entry, the illustrated Agreement (Figure 3.4c) introduces collection, storage, access and deletion, with optional details and access to the full policy. P16 preferred “short text with simple pictures or icons.” This supports exploring visual explanations, while Schaub et al., (2015) support considering notice timing and presentation. Neither preference nor completing the Agreement proves understanding. The supplied current screens for collection, optional cloud storage, access conditions, deletion and confirmation are documented in Appendix G.

Starting with a question lets users approach a specific concern without first reading a full dashboard. Keeping Access History within Events also retains the existing main navigation. A static FAQ could explain concepts but would provide a weaker connection to the relevant action. Putting all guidance inside eufy could reduce switching between websites: five of the six additional interview accounts supported an in-app privacy entry (Figure 3.2). The two-part design keeps explanations outside the device interface and decisions within it, but introduces an extra transition. Destination labels make that transition visible; they do not prove that users understand it.

The implemented prototype includes prepared search responses, links, visual explanations, sample access records and interactive permission choices. Viewer identities and visits are examples; masking and verification are demonstrations. Settings are stored in the browser and do not control a real account, camera or cloud service. Highlighted, step-by-step tutorials remain unimplemented. The Build team’s first evaluation priority is to observe whether users can complete this journey and explain the result: where a link leads, how past access differs from current permission, and which future access a change affects. These remain questions for evaluation, rather than benefits already demonstrated. Viewer verification is an exploratory interaction rather than an interview-validated requirement. It illustrates the distinction between identity checking and authorisation; its necessity and non-biometric alternatives require evaluation before any deployment.

3.5 What Changed Across the Iterations and Why

Table 3 summarises the main changes across the iterations. The design moved from presenting privacy information towards connecting that information with settings, permissions and records. Each change responds to identified evidence, but none has yet been evaluated in the connected web prototype.

Table 3. Changes across the iterations and their validation status.

Earlier problem or assumption	Final change	Supporting evidence	Validation status
Conditional privacy statements were too broad to interpret	Explanations distinguish storage locations, access roles and conditions, supported by data-flow views	Interim label testing; Finding 1	Not yet tested in the web prototype
Score meanings and opposite directions required interpretation	Scores labelled as illustrative, with factors and evidence available on demand	Round 2 testing; Finding 2	Post-guide clarity and trust were reported in the records; repeated exposure prevents causal attribution. Web version untested.
Understanding a product did not lead users to its settings	Question-based AIRIS answers with “Continue in eufy” links; tutorials planned	Findings 1 and 3	Not yet tested; tutorials unimplemented
Sharing did not show what others could see	Viewing Permissions with per-camera visibility options and consequence explanations	Finding 4; Geeng and Roesner (2019)	Not yet tested

Earlier problem or assumption	Final change	Supporting evidence	Validation status
Example access records offered no follow-up action	Expandable Access History linked to permissions, separating past and current access	Finding 5; early interviews (P3)	Not yet tested; login records not covered
Universal masking reduced the usefulness of footage	Masking offered as a choice within specific access situations	Interim critique (Section 3.3.3)	Not yet tested; masking accuracy unverified

The most significant change is the shift from a single information label to a connected journey. It addresses the gap between understanding and action identified in Findings 1 and 3, but introduces a transition between AIRIS and eufy that some participants' in-app preferences call into question. These changes therefore define what the Build team should evaluate (Section 5) rather than demonstrate improved understanding.

4 Ethical Considerations

4.1 Household Authority and Bystander Privacy

Home CCTV systems affect people beyond the account owner. Geeng & Roesner, (2019) describe how device installers often hold greater knowledge and control than other household members. Our owner-focused design could reinforce this imbalance if owners alone determine acceptable recording and access. Co-residents and visitors may be recorded without an account through which to inspect or challenge those decisions.

The permission interface should therefore explain who can view each camera and distinguish viewing from managing access. Permission changes should show their consequences before confirmation. However, an owner's confirmation cannot represent agreement from everyone being recorded. A future deployment would also need ways to communicate recording practices and receive concerns from people without accounts.

The proposed masking options introduce another boundary: hiding a face or person in a shared view does not necessarily remove that information from the original recording. The interface should state whether originals remain, who can access them and whether masking applies to playback or downloads. The current prototype demonstrates these choices without establishing technical protection of real footage.

Security and privacy may require different arrangements across households. Our design should support those decisions through visible permissions and clear limitations, while avoiding claims that masking alone resolves bystander privacy.

4.2 Trustworthy Guidance and Informed Choices

AIRIS should link explanations to sources and identify product, configuration and date conditions. For example, eufy describes local recording while noting cloud processing for optional thumbnail notifications; it also distinguishes default transport encryption from optional Advanced Encryption (eufy, n.d.-b). AIRIS should preserve these conditions and label unknowns, rather than turn a manufacturer statement into an independently verified assurance.

AIRIS also explains its own operation: the Privacy screen states that answers come from a curated question library, optional history stays in the browser, and AIRIS cannot read real eufy accounts or change device settings (Figure F3). The settings screen offers reduced animation and shows controls for optional search history and data export. These are visible prototype features, not a verified security or accessibility assessment.

Any retained privacy scores should be labelled as research-prototype summaries, with their factors and limitations available for inspection. They must not resemble official certification or imply independent security

testing. AIRIS's position outside the manufacturer also does not establish greater accuracy or trustworthiness. Its role and lack of manufacturer endorsement should be clear. Schaub et al., (2015) emphasise designing privacy notices around audiences, context and system constraints. We apply this through short explanations with optional detail and guidance near relevant decisions.

The agreement prototype's scrolling, checkboxes and waiting period indicate interaction, not understanding. The latest screen provides Disagree and Agree & Continue actions, keeps cloud recording optional and states that no legal consent is created by the demonstration. Evaluation should test whether declining and exiting work as expected, whether optional sharing remains separate, and whether users can explain the consequences before deciding.

4.3 Privacy Risks of Access Records and Viewer Verification

Access History may improve visibility while creating another source of household monitoring. Timestamps and viewing events could reveal when members are awake, away or checking particular spaces. Multi-user smart-home research highlights the need to consider how access controls and activity visibility affect household relationships (Zeng & Roesner, 2019).

A future implementation should collect only events needed for a defined purpose. Login events, permission changes and footage viewing should remain distinct, since one does not prove another. Shared members should know what is recorded and who can inspect it. Log access, retention and export permissions require evaluation rather than automatically granting owners unlimited access. Prototype records are examples, not evidence of actual surveillance.

Viewer verification currently simulates a process rather than establishing real biometric identity. Introducing face verification would create questions about capture, storage, deletion, unauthorised reuse and recognition failures. It could also exclude users unable or unwilling to complete the process.

Before introducing biometrics, we would need to establish why ordinary account authentication is insufficient and provide a non-biometric alternative. If viewer verification uses a phone or computer camera, it should explain why before requesting separate camera permission. Users should be able to refuse or cancel. The interface must say whether images leave the device, whether images or templates are retained, and who can access them. Capture should stop when verification ends. Browser camera permission is not consent to store biometric data or authorisation to view home CCTV.

5 Evaluation and Build Handover

5.1 Completed Evaluation and Its Limitations

The interim label testing (Rounds 1 and 2) is described in Section 3.1. It informed the Rating Label and its explanations, but its conclusions are limited. Round 1 included factual questions and preference measures, but preference did not establish superior comprehension; Round 2 involved repeated exposure and a small sample; and both concerned label interpretation rather than the complete web journey. The two rounds should not be treated as twelve unique participants without confirming participant overlap. Some Round 1 summary measures differ from the detailed records; Appendix B4 identifies these discrepancies. Numerical comparisons remain provisional pending reconciliation. See the interim test summaries in Appendices B and C.

Six additional interviews, **P13–P18**, informed the later requirements. P16 stated, "*I'm not always sure what other people can see or change,"* while P15 preferred privacy features *"in the same CCTV app instead of using a separate app."* These responses support clearer permissions and accessible guidance, while questioning the need for a separate platform. Because questions differed between interview rounds, we report themes rather than treating responses as a uniform survey.

No completed user testing of the connected AIRIS–eufy website is reported here. Clickable controls demonstrate proposed interactions but do not establish understanding of interface transitions, masking options or permission consequences. Demonstration records also cannot verify actual tracking or technical privacy protection. The following evaluation plan addresses these unresolved questions.

5.2 Evaluation Tasks and Success Criteria

The next evaluation should recruit household CCTV owners who manage shared access but have varying familiarity with privacy settings. Shared-access members should also be included where tasks concern their permissions and activity visibility. Sessions should use fictional accounts and footage, avoiding participants' actual household recordings. A consistent task protocol should combine observed interaction with participants' explanations. Instructions should describe a goal without naming the button or page to use. Initial attempts should be recorded before assistance; completion after a hint should remain distinguishable from independent completion.

Table 4. Evaluation tasks and evidence to collect.

Task	Evidence to collect
Find where footage storage is explained and locate a relevant setting	Navigation route, independent completion, wrong destinations and help required
Explain local storage and remote viewing	Whether participants distinguish storage location from transmission and identify configuration-dependent conditions
Identify a shared member's effective access	Correct identification of the camera, viewing option and management restrictions
Change a member's permission and explain the consequences	Successful change; understanding of future access, masking and existing downloaded copies
Interpret example activity events	Distinction between failed verification, successful access and recording history; recognition of what each event cannot prove

Across tasks, we will record completion, explanation accuracy, critical errors, assistance, time and confidence. Time will identify difficult steps rather than serve as evidence of understanding. Confidence will be compared with accuracy to identify confident misunderstandings.

For an initial formative study of eight participants, **provisional targets** are at least six independent completions per task and at least six correct explanations of its essential consequences. These are proposed design targets, not achieved results or population estimates. Any error suggesting that permission revocation deletes downloaded footage, masking destroys originals, or failed verification proves successful access should trigger revision and retesting. AIRIS should also be compared with an equivalent in-app explanation route. This would examine whether the additional interface contributes useful understanding or introduces unnecessary effort.

5.3 Functional and Non-Functional Requirements

The requirements below describe the intended future experience rather than capabilities already implemented. AIRIS can be developed as an independent guidance platform. However, changes to the official eufy interface, device permissions and access records would require collaboration with eufy and appropriate manufacturer support. The Build team can refine and evaluate these concepts but cannot independently implement them within the official app.

"Must" indicates a requirement necessary for the relevant component to function credibly. It does not imply that manufacturer cooperation has been secured.

Table 5. Functional and non-functional requirements.

Requirement	Responsibility or dependency	Priority	Verification
Functional: Explain storage, transmission and access conditions for supported configurations	AIRIS; requires verifiable product information	Must	Check explanations against documentation and configuration scenarios
Functional: Guide users towards relevant official settings and identify service transitions	AIRIS; depends on available eufy settings	Must	Observe navigation tasks and check whether users recognise the official destination
Functional: Show effective permissions for each member and camera	Proposed eufy redesign; requires manufacturer cooperation	Must	Test the concept against permission scenarios; verify actual behaviour if implemented
Functional: Explain permission changes, masking limitations and existing copies	Proposed eufy redesign; requires confirmed technical behaviour	Must	Test consequence understanding and compare explanations with implementation
Functional: Provide appropriately scoped footage-access records; investigate login history separately	Requires manufacturer access to reliable event data	Should	Compare displayed records with controlled events and document coverage gaps
Non-functional: Provide inspectable sources, score meanings and uncertainty	AIRIS	Must	Audit claims, source relevance and illustrative score labels
Non-functional: Support understandable and accessible interaction	Both proposed components	Must	Comprehension tasks, keyboard navigation and assistive-technology checks
Non-functional: Distinguish guidance, simulation and confirmed actions	AIRIS and the demonstration prototype	Must	Check that users recognise which actions affect real devices
Non-functional: Minimise activity data and restrict visibility and retention	Any future logging implementation	Must	Review collected fields, access roles and retention behaviour

Without manufacturer cooperation, the deliverable should remain an independent AIRIS platform and a clearly identified eufy concept demonstration. AIRIS must not report that a real permission has changed unless a supported integration confirms it.

5.3.1 Intended Use Cases

These intended outcomes require evaluation. Actions in the eufy redesign remain simulated until a manufacturer-supported implementation is available.

UC1 Understand storage. A CCTV owner asks where recordings go. AIRIS explains the selected model's local and optional cloud paths and links to the relevant setting. Missing configuration evidence is identified as unknown.

UC2 Restrict sharing. An owner reviews a member's camera access and changes the viewing scope. The interface explains future effects and warns that downloaded copies remain. A failed save must not appear successful.

UC3 Review access. An owner inspects who viewed footage and when, then opens permissions. Viewing, recording and login events remain distinct. Missing records must not be presented as proof of no access.

UC4 Decide on backup. An owner reviews a cloud-backup request's purpose and visibility, then allows or declines it. Declining leaves the task unapproved; backup approval does not grant a person viewing access.

5.4 Project Risks and Prioritised Questions for the Build Team

The main implementation risk is dependence on a manufacturer whose interface and device behaviour our team cannot directly change. The Build team should resolve this dependency before treating the redesigned eufy experience as an implementable system.

Priority 1: Is manufacturer collaboration feasible?

The proposed permission interface and Access History require eufy's cooperation, technical support and agreement on responsibilities. The team should investigate potential collaboration and available integration mechanisms. Required evidence includes confirmation of supported capabilities, accessible event types and implementation responsibilities. Public documentation alone should not be treated as permission or assurance that the proposed redesign can be adopted. Without cooperation, the scope should remain AIRIS guidance and a simulated eufy demonstration.

Priority 2: Does AIRIS provide value through currently available settings?

AIRIS should remain useful without assuming that eufy adopts our redesign. The team should compare question-based guidance with an equivalent in-app information route, using settings available in the official app. Evaluation should examine understanding, navigation, switching effort and recognition of service boundaries. It should establish whether AIRIS contributes useful support and whether users mistakenly believe it is operated or endorsed by eufy.

Priority 3: Can proposed controls and records be implemented accurately?

If collaboration becomes feasible, the team should investigate permission enforcement, masking behaviour and event coverage through controlled technical trials. Required evidence includes whether restrictions apply across access routes, whether originals remain available, and which events can be reliably attributed. Login, viewing and failed verification must remain distinct. Unsupported capabilities should be removed or explicitly limited rather than presented as complete protection.

Priority 4: Can guidance remain accurate within a manageable scope?

Official interfaces and product configurations may change, making instructions outdated. The team should prioritise a limited set of supported configurations, record source-review dates and test how unavailable or outdated guidance is communicated. Evidence should demonstrate that affected content can be identified and corrected. This approach keeps the proposal useful while avoiding dependence on a broad redesign that has neither manufacturer agreement nor technical validation.

The handover therefore offers two possible paths: develop and evaluate independent guidance using existing capabilities, or investigate the integrated redesign through a future manufacturer partnership.

6 Team Process and Critical Reflection

6.1 Project Milestones and Changes to the Original Plan

Our earlier work focused on communicating CCTV privacy information through paper labels and alternative visual representations. During this stage, we expanded that focus towards an interactive experience connecting explanations with settings and access management. The main milestones reflect this shift.

Table 6. Project milestones and changes to the original plan.

Earlier direction	Work completed and change
Compare privacy label formats	Interim label testing (Rounds 1 and 2) identified unclear access conditions and difficulties interpreting ratings.
Understand CCTV privacy concerns	Six additional interviews explored everyday app use, sharing and settings.
Develop privacy information displays	The web prototype connected AIRIS explanations with proposed eufy permission and activity-history interactions.
Prepare the final proposal	Prototype development, reporting and exhibition materials progressed in parallel.

This change broadened what we could demonstrate, but also increased the number of assumptions requiring evaluation. Paper labels allowed focused comparisons of information presentation; the website introduced navigation, permissions and relationships between interfaces. Results from the earlier tests therefore cannot establish that the complete updated experience works.

The current prototype demonstrates key interactions rather than a production system. Step-by-step tutorials remain proposed, while real device integration and technical enforcement fall outside the demonstrated implementation. Recognising these boundaries helped us distinguish design progress from evidence of effectiveness.

6.2 How Feedback Changed Our Decisions

A central weakness in our previous report was the lack of interview quotations supporting our interpretations. Readers could see our conclusions but could not easily trace them to participants' responses. This made the relationship between research and design less convincing.

In response, we revised the findings to include English interview quotations with participant identifiers. We also distinguished participant statements, testing observations and our own interpretations. The additional affinity map makes thematic grouping visible, while the requirements table connects findings with proposed design responses. Internal links connect selected claims with supporting quotations and test records in the appendices.

This process changed how we assessed evidence strength. For example, P15's request supports exploring login history, but one household account cannot establish that all households need it; viewing records rest instead on earlier evidence about footage access. Similarly, preferences for privacy features within an existing app challenge our decision to introduce AIRIS. We retained these limitations instead of selecting only evidence favourable to the proposal.

Traceability still requires careful checking. Quotations must match their source documents, and references must follow the final Finding numbering. Adding a quotation or link improves accountability, but does not resolve limited stakeholder coverage or validate an untested feature.

6.3 Collaboration Strengths, Challenges and Lessons

Clear task allocation helped us progress efficiently. Everyone contributed to part of the prototype, while exhibition and reporting tasks proceeded alongside development. For example, one member could prepare the poster while others worked on the report and promotional leaflet. This allowed several deliverables to advance without waiting for one task to finish.

GitHub supported shared prototype development, while Google Docs made report contributions and progress visible. We used WeChat to confirm changes, discuss specific questions and coordinate improvements. Meetings provided opportunities to clarify problems and redistribute work when needed. These tools served different purposes: shared files held the developing work, messaging supported quick decisions, and meetings helped resolve questions requiring fuller discussion.

However, completing separate contributions is not the same as producing a coherent proposal. Parallel work creates a need to check that feature descriptions, research claims and prototype behaviour remain consistent. Our changing Finding structure illustrates this: revising the findings also required updates to the requirements table, cross-references and affinity-map caption. A locally correct section could still conflict with another part of the report.

We would retain the clear division of work and visible shared progress. We would strengthen integration by assigning responsibility for a final consistency review and maintaining one agreed record of current features, evidence and outstanding questions. Before submission, the team should check quotations, references and implementation status together. This would preserve the speed of parallel work while making the final argument easier to follow.

6.4 Final Design Position

Our proposal pairs AIRIS guidance with a conceptual eufy redesign for permissions and activity records. Interviews and label tests support clearer explanations and visible access boundaries. They do not validate the separate platform, technical protection or effects on behaviour. The Build team should first test the connected journey and clarify manufacturer dependencies, record coverage, masking and verification.

References

Emami-Naeini, P., Agarwal, Y., Faith Cranor, L., & Hibshi, H. (2020). Ask the Experts: What Should Be on an IoT Privacy and Security Label? *2020 IEEE Symposium on Security and Privacy (SP)*, 447–464. https://doi.org/10.1109/SP40000.2020.00043

Farke, F. M., Balash, D. G., Golla, M., Dürmuth, M., & Aviv, A. J. (2021). Are Privacy Dashboards Good for End Users? Evaluating User Perceptions and Reactions to Google's My Activity (Extended Version). *arXiv.Org*. https://www.proquest.com/docview/2535628918?pq-origsite=primo&searchKeywords=Are%20privacy%20dashboards%20good%20for%20end%20users%3F%20Evaluating%20user%20perceptions%20and%20reactions%20to%20Google%E2%80%99s%20My%20Activity.%20In%20Proceedings%20of%20the%2030th%20USENIX%20Security%20Symposium%20(pp.%20483%E2%80%93500).&sourcetype=Working%20Papers

Feng, Y., Yao, Y., & Sadeh, N. (2021). A Design Space for Privacy Choices: Towards Meaningful Privacy Control in the Internet of Things. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, 1–16. https://doi.org/10.1145/3411764.3445148

Geeng, C., & Roesner, F. (2019). Who's In Control?: Interactions In Multi-User Smart Homes. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 1–13. https://doi.org/10.1145/3290605.3300498

Lovejoy, B. (2022, December 21). *Eufy camera security breach admission leaves many questions unanswered*. https://9to5mac.com/2022/12/21/eufy-camera-security-breach/

Lovejoy, B. (2023, February 1). *Anker admits to lying about Eufy security camera encryption; describes future plans*. http://9to5mac.com/2023/02/01/eufy-security-camera-encryption/

Manandhar, S., Kafle, K., Andow, B., & Nadkarni, A. (2022). *Smart Home Privacy Policies Demystified: A Study of Availability, Content, and Coverage*.

Marky, K., Prange, S., Mühlhäuser, M., & Alt, F. (2021). Roles Matter! Understanding Differences in the Privacy Mental Models of Smart Home Visitors and Residents. *20th International Conference on Mobile and Ubiquitous Multimedia*, 108–122. https://doi.org/10.1145/3490632.3490664

Schaub, F., Balebako, R., Durity, A. L., & Cranor, L. F. (2015). *A Design Space for Effective Privacy Notices*. 1–17. https://www.usenix.org/conference/soups2015/proceedings/presentation/schaub

Windl, M., Thalhammer, P., Müller, D., Schmidt, A., & Feger, S. S. (2025). PrivacyHub: A Functional Tangible and Digital Ecosystem for Interoperable Smart Home Privacy Awareness and Control. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 1–15. https://doi.org/10.1145/3706598.3713517

Zeng, E., & Roesner, F. (2019). *Understanding and Improving Security and Privacy in Multi-User Smart Homes: A Design Exploration and In-Home User Study*. 159–176. https://www.usenix.org/conference/usenixsecurity19/presentation/zeng

Zheng, S., Apthorpe, N., Chetty, M., & Feamster, N. (2018). User Perceptions of Smart Home IoT Privacy. *Proceedings of the ACM on Human-Computer Interaction*. https://dl.acm.org/doi/10.1145/3274469

Appendices

Appendix A - Interview Materials and Evidence Index

A1 Interview record index

Interview IDs identify records, not test sessions. R1-Pxx and R2-Pxx identify label-test records separately. P13 is the store participant; P14 and P18 are household participants. Earlier P15–P20 labels in working notes are superseded by P13–P18.

Table A1. Interview records and source documents.

ID	Research stage	Context	Source document
P1	Early interviews	Early interview record	interviews.docx
P2	Early interviews	Early interview record	interviews.docx
P3	Early interviews	Prospective buyer	interviews.docx
P4	Early interviews	Early interview record	interviews.docx
P5	Early interviews	Early interview record	interviews.docx
P6	Early interviews	Early interview record	p6-8.docx
P7	Early interviews	Early interview record	p6-8.docx
P8	Early interviews	Early interview record	p6-8.docx
P9	Early interviews	Early interview record	p9-10.docx
P10	Early interviews	Early interview record	p9-10.docx
P11	Early interviews	Installer perspective	p11-12.docx
P12	Early interviews	Early interview record	p11-12.docx
P13	Additional interviews	Store CCTV use	CCTV_Interview_13_Transcript.docx; P13.txt
P14	Additional interviews	Household CCTV use	CCTV_Interview_14_Transcript.docx; P14.txt
P15	Additional interviews	Household CCTV use	CCTV_Interview_15_Transcript.docx
P16	Additional interviews	Household CCTV use	CCTV_Interview_16_Transcript.docx
P17	Additional interviews	Household CCTV use	CCTV_Interview_17_Transcript.docx
P18	Additional interviews	Household CCTV use	CCTV_Interview_18_Transcript.docx; P18.txt

A2 Additional interview guide

Source: Interview2_Question.docx. The following are the 14 English core questions from the supplied semi-structured guide. Follow-up prompts were flexible; not every question was asked in every interview. Although the guide targeted household owners, P13 concerned a store and is analysed separately.

Q1. What CCTV brand do you currently use, and how long have you been using it?

Q2. What functions do you use most often in your CCTV app?

Q3. Can you describe how you normally use your CCTV app, from opening it to finishing your task?

Q4. What do you like most about your current CCTV app?

Q5. What do you dislike or find frustrating about your current CCTV app?

Q6. What improvements would you like to see in your current CCTV app, especially in its settings and functions?

Q7. Have you ever looked for privacy-related information in your CCTV app?

Q8. Have you ever changed any privacy or security settings in your CCTV app?

Q9. Have you ever needed to check or manage who can access your CCTV camera or recordings?

Q10. What notifications or activity information does your CCTV app currently provide?

Q11. Have you ever noticed anything unusual or unexpected when using your CCTV app?

Q12. Based on your experience, what are the three most important things a good CCTV app should provide?

Q13. If you wanted to understand or manage your CCTV privacy better, how would you prefer to access these functions?

Q14. How would you prefer privacy information and consent requests to be presented in your CCTV app?

The guide instructed interviewers to obtain consent before recording, avoid passwords, account details and private footage, and ask for specific experiences before suggestions. These are protocol instructions, not confirmation that every procedure was documented in every session.

A3 Quotations and source locations

The index gives the supplied wording, record ID and source location for each extract. Selected direct quotations and researcher observations are distinguished below. The source documents identify which records contain edited summaries; supplementary summaries are not additional participant statements.

R1-P02 — Finding 1

“Ah, what do you mean by "for a specific purpose"?”

Source: R1-P2.txt; line 98.

P3 — Finding 5

“I'd want to know who accessed it, when they accessed it and why.”

Source: interviews.docx; question about company access.

P15 — Findings 1, 3

“I want to know if the camera company can see my recordings. But the privacy information wasn't very easy to find.”

Source: CCTV_Interview_15_Transcript.docx; main Q7.

P16 — Finding 3

“I looked in the settings, but I couldn't find a clear answer. There was a lot of text, so I didn't read everything.”

Source: CCTV_Interview_16_Transcript.docx; main Q7.

P16 — Finding 4

“but I'm not always sure what other people can see or change. I wish the app explained this more clearly.”

Source: CCTV_Interview_16_Transcript.docx; main Q9, continuous excerpt.

P17 — Finding 4

“I think it would be useful to have a clear page showing who can see the camera.”

Source: CCTV_Interview_17_Transcript.docx; main Q9, continuous excerpt.

P15 — Finding 5

“I would also like to see login history.”

Source: CCTV_Interview_15_Transcript.docx; main Q10.

P15 — Finding 3; AIRIS limitation

“I prefer to have all privacy features in the same CCTV app instead of using a separate app.”

Source: CCTV_Interview_15_Transcript.docx; main Q13.

P16 — Finding 3

“It would also be helpful if I could click to read more details when I want to.”

Source: CCTV_Interview_16_Transcript.docx; main Q14.

P17 — Everyday-use context

“If everything looks fine, I will close the app.”

Source: CCTV_Interview_17_Transcript.docx; main Q3.

P13 — Store context; access visibility

“I mainly want to understand who could remotely access the cameras and recordings.”

Source: P13.txt; 04:00.

P13 — Activity-history exploration

“A clearly, a clear activity history would be helpful.”

Source: P13.txt; 03:13.

R2-P06 — Score direction observation

Researcher observation: the participant identified Exposure as lower-is-preferable and User Control as higher-is-preferable, while describing initial hesitation about the familiar assumption that higher scores are better. The record does not show a sustained incorrect answer.

Source: A1-2/evidence/R2-P6.txt, 04:15–05:04, lines 17–23. Related prepared record: Round2_P6_Test_Result.docx, Stage 1 Q3.

A4 Selected additional interview records

The following extracts retain complete question-and-answer pairs from the supplied documents. Each source note identifies the status of the record. These extracts provide context beyond the short quotations used in the report.

P13 Interview record extracts

Source: CCTV_Interview_13_Transcript.docx; prepared interview record. Original source: P13.txt.

Interviewer: Have you ever looked for privacy-related information in your CCTV app?

Participant: Yes. Because the cameras record customers, staff, and reptile areas, I have checked account security and device-sharing settings. I want to know who can remotely access cameras and recordings.

Interviewer: Have you ever needed to check or manage who can access your cameras or recordings?

Participant: Yes. Staff sometimes need access to cameras covering the main store or entrance, but I do not want everyone to access every camera or change device settings. I would prefer to manage permissions in one place.

Interviewer: What notifications or activity information does your CCTV app currently provide?

Participant: I receive device and event notifications, including when a camera goes offline. I would also like a clear record of account logins, video access, and settings changes.

Interviewer: How would you prefer to access CCTV privacy functions?

Participant: I would prefer them within my existing CCTV app. Ideally, one privacy or security page would show user permissions, remote access, connected devices, and recent activity. I would consider another tool only if it were secure and trustworthy.

P14 Interview record extracts

Source: CCTV_Interview_14_Transcript.docx; prepared interview record. Original source: P14.txt.

Interviewer: Have you ever looked for privacy-related information in your CCTV app?

Participant: I have seen the settings, but I have not thought much about whether other people can see the footage. When setting up the camera, I understood that recordings were saved locally. The settings seem fairly simple.

Interviewer: Have you ever needed to check or manage who can access your cameras or recordings?

Participant: Two people in our household share access to the CCTV view. We have not given anyone else permission.

Interviewer: What notifications or activity information does your CCTV app currently provide?

Participant: It sends a notification when movement is detected. I can tap the notification to see the recording from that time.

Interviewer: How would you prefer to access CCTV privacy functions?

Participant: I think privacy depends partly on how much the company can be trusted. I would like a simpler interface and strong protection against privacy leaks. I did not specify an exact location for these controls.

P15 Interview record extracts

Source: CCTV_Interview_15_Transcript.docx; prepared interview record. Unclear markers and the supplied wording are retained.

Interviewer: Have you ever looked for privacy-related information in your CCTV app?

Participant: Yes, I tried to find information about how my videos are stored and who can view them. I want to know if the camera company can see my recordings. But the privacy information wasn't very easy to find.

Interviewer: Have you ever needed to check or manage who can access your CCTV cameras or recordings?

Participant: Actually, I think my CCTV doesn't have this function, but I invited my parents too. They also can see the CCTV.

Interviewer: What notifications or activity information does your CCTV app currently provide?

Participant: My app sends motion detection alerts, and it shows recording history. These are useful, but sometimes I get too many notifications. I would also like to see login history.

Interviewer: If you wanted to understand or manage your CCTV privacy better, how would you prefer to access these functions?

Participant: I prefer to have all privacy features in the same CCTV app instead of using a separate app. A clear privacy section with simple buttons would make it easier to manage.

P16 Interview record extracts

Source: CCTV_Interview_16_Transcript.docx; prepared interview record. Unclear markers and the supplied wording are retained.

Interviewer: Have you ever looked for privacy-related information in your CCTV app?

Participant: Yes, I tried to find out where my videos are stored. I looked in the settings, but I couldn't find a clear answer. There was a lot of text, so I didn't read everything. I think this information should be easier to understand.

Interviewer: Have you ever needed to check or manage who can access your CCTV camera or recordings?

Participant: Yes, I wanted to let one of my family members use the camera. I found the [unclear] and added them, but I'm not always sure what other people can see or change. I wish the app explained this more clearly.

Interviewer: What notifications or activity information does your CCTV app currently provide?

Participant: The app sends me notifications [unclear] movement. I can also check past videos. I think these features are useful, but I sometimes get too many alerts. I don't usually check the activity [unclear].

Interviewer: If you wanted to understand or manage your CCTV privacy better, how would you prefer to access these functions?

Participant: I would prefer to have all privacy settings in one place. Maybe there could be a privacy button on the main page. I think it is better to have these features in the same app because I don't want to download another app.

Interviewer: How would you prefer privacy information and the consent request to be presented in your CCTV app?

Participant: I prefer short text with simple pictures or icons. Usually, I skip privacy information because it is too long and hard to read. I think pictures can help me understand the main points faster. It would also be helpful if I could click to read more details when I want to.

P17 Interview record extracts

Source: CCTV_Interview_17_Transcript.docx; prepared interview record. Unclear markers and the supplied wording are retained.

Interviewer: Have you ever looked for privacy-related information in your CCTV app?

Participant: Not really. I know there are some privacy settings in the app but I have never really spent time reading them and I usually just adjust the camera function but I do care about privacy especially who can see my camera and where my video are stored.

Interviewer: Have you ever needed to check or manage who can access your CCTV camera or recordings?

Participant: Not very often. Usually only my family members use the camera so I haven't really needed to change who has access but I think it would be useful to have a clear page showing who can see the camera.

Interviewer: What notifications or activity information does your CCTV app currently provide?

Participant: My app mainly sends notification when it detects movement and sometimes they are useful especially when nobody is home but sometimes I got too many notifications for small things.

Interviewer: If you want to understand or manage your CCTV privacy better, how would you prefer to access these functions?

Participant: I would prefer to have everything in the same app rather than using another app. Maybe there could be a privacy section on the main page or in the settings. I would like to see who can access my camera, where my video is stored and how I can change these settings. I think simple explanations would also be helpful because not everyone understands technical terms.

P18 Interview record extracts

Source: CCTV_Interview_18_Transcript.docx; prepared interview record. Original source: P18.txt.

Interviewer: Have you ever looked for privacy-related information in your CCTV app?

Participant: I have opened a privacy notice, but I have not actively looked for much more information.

Interviewer: Have you ever needed to check or manage who can access your cameras or recordings?

Participant: Yes. Only my parents and I should have access. To add someone, they install the same app, search for the shared camera name, and enter a password.

Interviewer: What notifications or activity information does your CCTV app currently provide?

Participant: It provides alerts or records for unusual activity, such as someone walking through the living room at night.

Interviewer: How would you prefer to access CCTV privacy functions?

Participant: I would like privacy controls to be clearly visible on the app's main page, so they are easy to find and manage.

Interviewer: How should privacy information and informed consent be presented?

Participant: I would like a notice when the app is first used and when each camera is first activated. I would also like short explanations with text and images or icons.

A5 Selection and Interpretation of Early Interview Evidence

Figure 3.1 presents selected extracts rather than retrospective frequency counts. The available records do not establish a named coder or a complete participant-by-category matrix for the earlier totals. Those totals are therefore omitted. Extracts were selected for explicit relevance to finding privacy information or reviewing footage access; participant wording and design interpretation remain separate.

P3: "I'd want to know who accessed it, when they accessed it and why." Source: A1/interviews.docx, P3, company-access question.

P7: "Yes, absolutely. There should be some kind of record." Source: A1-2/p6-8.docx, Participant B, company-access question.

P9: "I would prefer a simple page in the app and a label that clearly shows where the videos are stored, how long they are kept, who can access them, and how to delete them." Source: A1-2/p9-10.docx, first interview, presentation-preference question.

Appendix B - Round 1 Label Comparison

Round 1 compared Facts Only (A), Rating Label (B) and XY Privacy Profile (C), using the same Ring product information. The following table reproduces the interim report's summary records; it is not a verbatim transcript or a new assessment of participant performance. Accuracy is reported on the recorded three-question scale; confidence uses a five-point scale; preference rank 1 indicates first choice.

Source: Studio3-Proposal-2026-Problem-ANN.docx, Appendix G. Original test materials remain separate supporting files.

B1 Method and tasks

The same Ring product information was shown as Facts Only (A), Rating Label (B) and XY Privacy Profile (C). Participants answered three factual questions concerning manufacturer access, user control and storage; the recorded outputs include accuracy, completion time and confidence. Format preference was recorded separately. Presentation order varied across the checkpoint records.

B2 Format comparison questions

Which version was easiest to understand? Why?
Which version helped you find information fastest?
What does “Data Access Exposure” mean to you?
What does “User Control” mean to you?
Was any information confusing?
Was any important privacy information missing?
Which version would you prefer to see before buying a CCTV camera? Why?

B3 Summary results

Table B1. Interim summary of Round 1 label comparison records.

Participant	Presentation Order	Accuracy A / B / C	Time A / B / C	Confidence A / B / C	Preference Rank A / B / C
R1-P01	B → C → A	2/3 / 3/3 / 3/3	62s / 50s / 42s	3/5 / 4/5 / 4/5	1 / 3 / 2
R1-P02	A → B; C also evaluated	2/3 / 2/3 / 2/3	2m / 2m / 1m30s	3/5 / 3/5 / 2/5	2 / 1 / 3
R1-P03	A → B → C	3/3 / 3/3 / 3/3	42s / 55s / 67s	5/5 / 4/5 / 3/5	1 / 2 / 3
R1-P04	B → C → A	2.5/3 / 3/3 / 3/3	58s / 30s / 39s	4/5 / 5/5 / 4/5	3 / 1 / 2
R1-P05	B → A → C	3/3 / 3/3 / 2/3	82s / 72s / 103s	4/5 / 5/5 / 3/5	2 / 1 / 3
R1-P06	C → B → A	3/3 / 3/3 / 2/3	69s / 58s / 111s	4/5 / 5/5 / 3/5	3 / 1 / 2

These records concern label interpretation. They do not demonstrate the usability, technical enforcement or effectiveness of the connected AIRIS–eufy website.

B4 Record reconciliation and interpretation limits

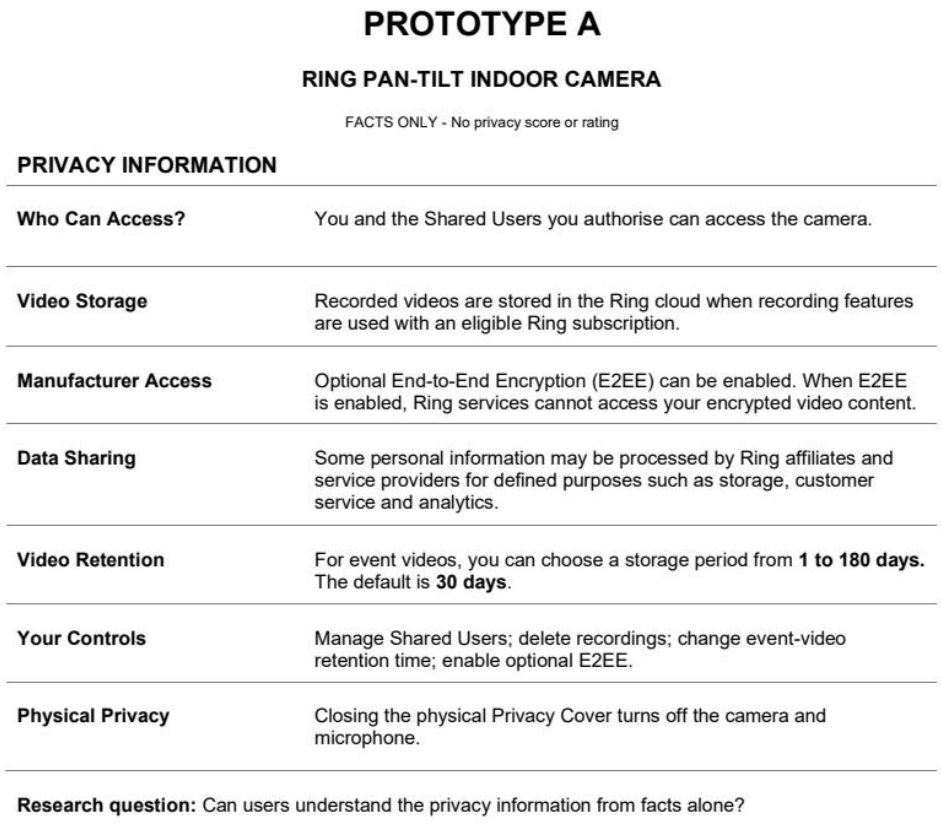
The interim summary and P1-2.docx detailed records do not agree on every Round 1 measure. They are retained as separate source versions rather than silently combined. The preference totals in the report are attributed to the checkpoint summary. Numerical performance comparisons should not be treated as reconciled results.

Table B2. Discrepancies between checkpoint summaries and detailed records.

Record	Checkpoint summary	Detailed record	Implication
R1-P01 accuracy A/B/C	2/3; 3/3; 3/3	2.5/3; 2.5/3; 2.5/3	Do not recompute an overall accuracy effect from mixed versions.
R1-P02 accuracy and C time	2/3 across formats; C 1m30s	2.5/3; 2.5/3; 2/3; C 1m36s	Scores and timing require source reconciliation.
R1-P02 preference	B ranked first	Final table: not recorded; narrative describes C preference	The reported 4/2/0 distribution is a checkpoint summary, not a verified reconstruction.

B5 Prototype materials used for format comparison

The following pages reproduce the earlier label material from Ring_Privacy_Prototype_ABC_Revised.pdf. Product descriptions and research ratings are historical study stimuli, not current product advice or official certification.



PROTOTYPE A

RING PAN-TILT INDOOR CAMERA

FACTS ONLY - No privacy score or rating

PRIVACY INFORMATION

Who Can Access?	You and the Shared Users you authorise can access the camera.
Video Storage	Recorded videos are stored in the Ring cloud when recording features are used with an eligible Ring subscription.
Manufacturer Access	Optional End-to-End Encryption (E2EE) can be enabled. When E2EE is enabled, Ring services cannot access your encrypted video content.
Data Sharing	Some personal information may be processed by Ring affiliates and service providers for defined purposes such as storage, customer service and analytics.
Video Retention	For event videos, you can choose a storage period from **1 to 180 days.** The default is **30 days**.
Your Controls	Manage Shared Users; delete recordings; change event-video retention time; enable optional E2EE.
Physical Privacy	Closing the physical Privacy Cover turns off the camera and microphone.

Research question: Can users understand the privacy information from facts alone?

Figure B1. Earlier label stimulus, original material page 1.

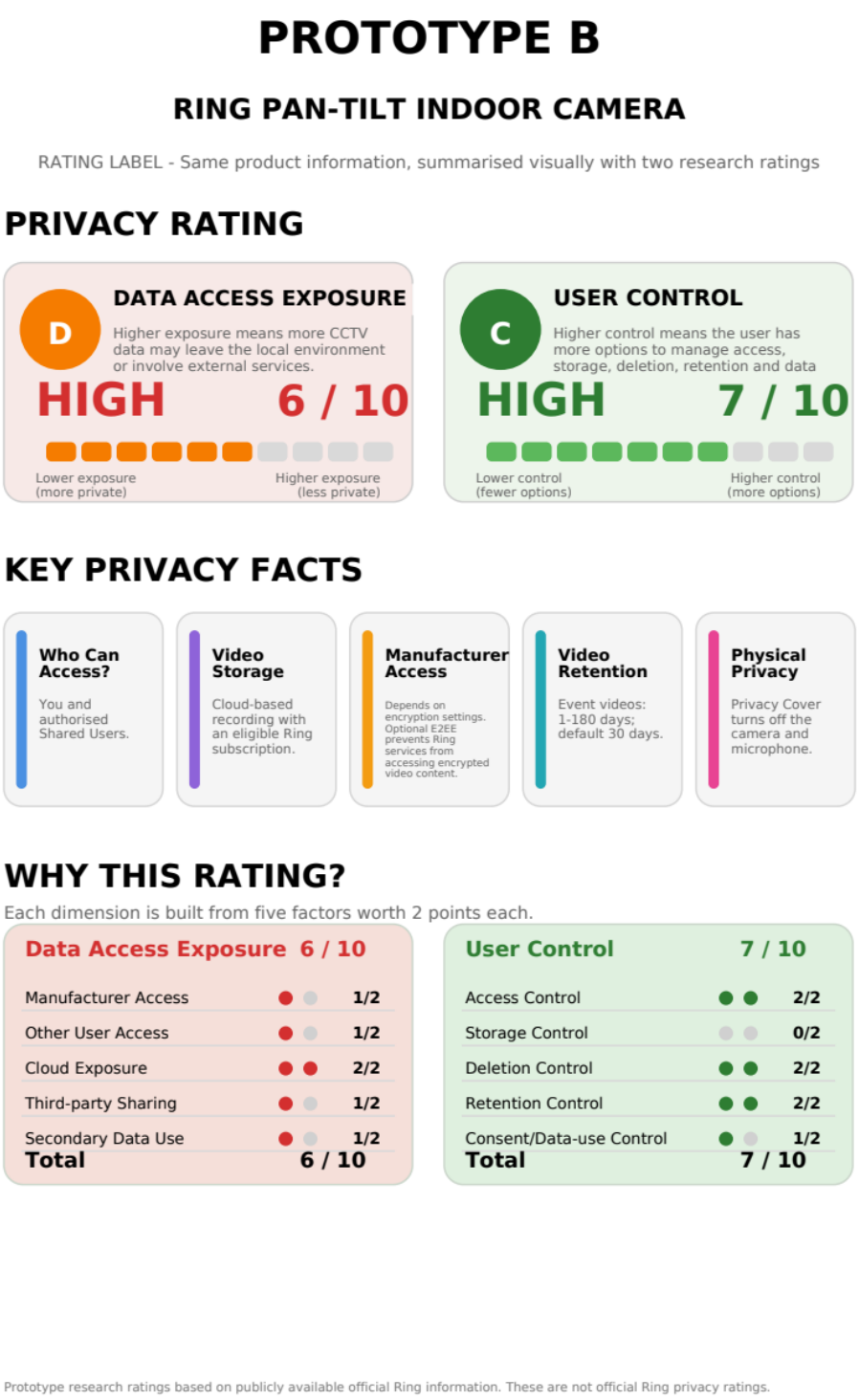
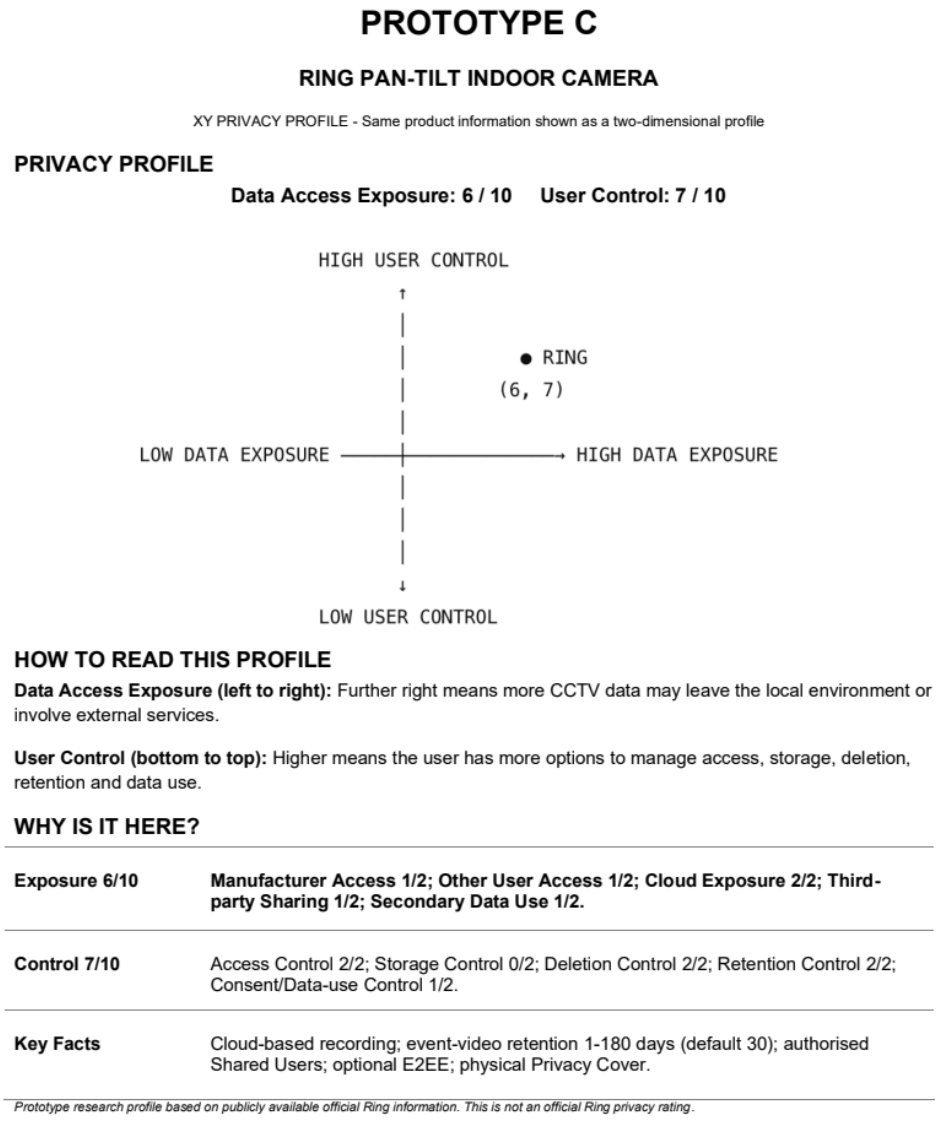


Figure B2. Earlier label stimulus, original material page 2.



PROTOTYPE C

RING PAN-TILT INDOOR CAMERA

XY PRIVACY PROFILE - Same product information shown as a two-dimensional profile

PRIVACY PROFILE

Data Access Exposure: 6 / 10 User Control: 7 / 10

HIGH USER CONTROL

● RING

(6, 7)

LOW DATA EXPOSURE → HIGH DATA EXPOSURE

LOW USER CONTROL

HOW TO READ THIS PROFILE

Data Access Exposure (left to right): Further right means more CCTV data may leave the local environment or involve external services.

User Control (bottom to top): Higher means the user has more options to manage access, storage, deletion, retention and data use.

WHY IS IT HERE?

Exposure 6/10	**Manufacturer Access 1/2; Other User Access 1/2; Cloud Exposure 2/2; Third-party Sharing 1/2; Secondary Data Use 1/2.**
Control 7/10	Access Control 2/2; Storage Control 0/2; Deletion Control 2/2; Retention Control 2/2; Consent/Data-use Control 1/2.
Key Facts	Cloud-based recording; event-video retention 1-180 days (default 30); authorised Shared Users; optional E2EE; physical Privacy Cover.

Prototype research profile based on publicly available official Ring information. This is not an official Ring privacy rating.

Figure B3. Earlier label stimulus, original material page 3.

Appendix C - Round 2 Cross Product Comparison

Round 2 compared labels for eufy, Ring and Tapo before and after participants reviewed the scoring guide and supporting evidence. The table reproduces the interim report's summary records. Before-and-after differences involve repeated exposure and should not be interpreted as a causal effect. Participant overlap between rounds is unconfirmed.

C1 Two-stage protocol

Stage 1: participants reviewed the three product labels without the scoring guide. Stage 2: the scoring rules and evidence sources were introduced before interpretation and choice were revisited. The supplied protocol requested varied product order; recorded orders should be used when analysing individual sessions.

C2 Core questions

Q1 When buying a home CCTV product, what information would you normally consider? Select the three most important items and explain why.

Q2 Please scan the three labels. What did you notice first?

Q3 What do Data Access Exposure and User Control mean to you? For each score, is higher or lower better?

Q4 Before seeing the scoring rules, can you understand this rating system? What is unclear, and how would you explain the label to another person?

Q5 What do Ring's 5-7/10, Tapo's 3-7/10, the question marks, and the outlined score segments mean to you?

Q6 Find the most important differences among the three products in Manufacturer Access, Video Storage, Video Retention, and Encryption.

Q7 If you had to choose one product using only the scores and information on these labels, which product would you choose and why?

Q8 Which three pieces of information had the greatest influence on your choice? Rank them in order of importance.

Q9 Which disadvantages would you accept, and which would be unacceptable?

Q10 How confident are you in your first product choice? Rate your confidence from 1 to 5 and explain your rating.

Q11 After reading the scoring guide, explain Data Access Exposure, User Control, score ranges, and question marks again. What changed in your understanding?

Q12 After reviewing the scoring rules and evidence sources, which product would you choose? Is this the same as your first choice?

Q13 Would you accept higher Data Access Exposure in exchange for stronger User Control, greater convenience, or other features? Why or why not?

Q14 Which part of the label was most helpful? Which part was hardest to understand or easiest to misinterpret?

Q15 If you could change only one part of the label, what would you change? How confident are you now in the rating system and your final product choice?

C3 Measures and recording

The protocol recorded initial comprehension, comparison accuracy out of four, comparison time, first attention, reading path, hesitation, misinterpretation, initial/final choice and confidence, and reported trust after evidence review. Observed clarification after the guide cannot separate the guide’s effect from repetition or moderator support.

Table C1. Interim summary of Round 2 cross-product comparison records.

Participant	Product Order	Comparison Accuracy	Comparison Time	Initial → Final Choice	Choice Confidence Before → After	Final System Confidence / Clarity	Reported Trust After Evidence
R2-P01	eufy → Ring → Tapo	4/4	3m06s	eufy → eufy	4/5 → 4/5	5/5	Higher
R2-P02	Ring → Tapo → eufy	3/4	5m18s	Ring → eufy	3/5 → 4/5	4/5	Higher
R2-P03	Ring → Tapo → eufy	4/4	2m18s	eufy → eufy	3/5 → 4/5	4/5	Higher
R2-P04	eufy → Ring → Tapo	4/4	2m20s	eufy → eufy	3/5 → 4/5	4/5	Higher
R2-P05	Ring → Tapo → eufy	3/4	2m45s	Tapo → Tapo	3/5 → 3/5	4/5	Same
R2-P06	eufy → Ring → Tapo	4/4	2m41s	eufy → eufy	4/5 → 5/5	4/5	Higher

These records concern label interpretation. They do not demonstrate the usability, technical enforcement or effectiveness of the connected AIRIS–eufy website.

Appendix D - Affinity Mapping

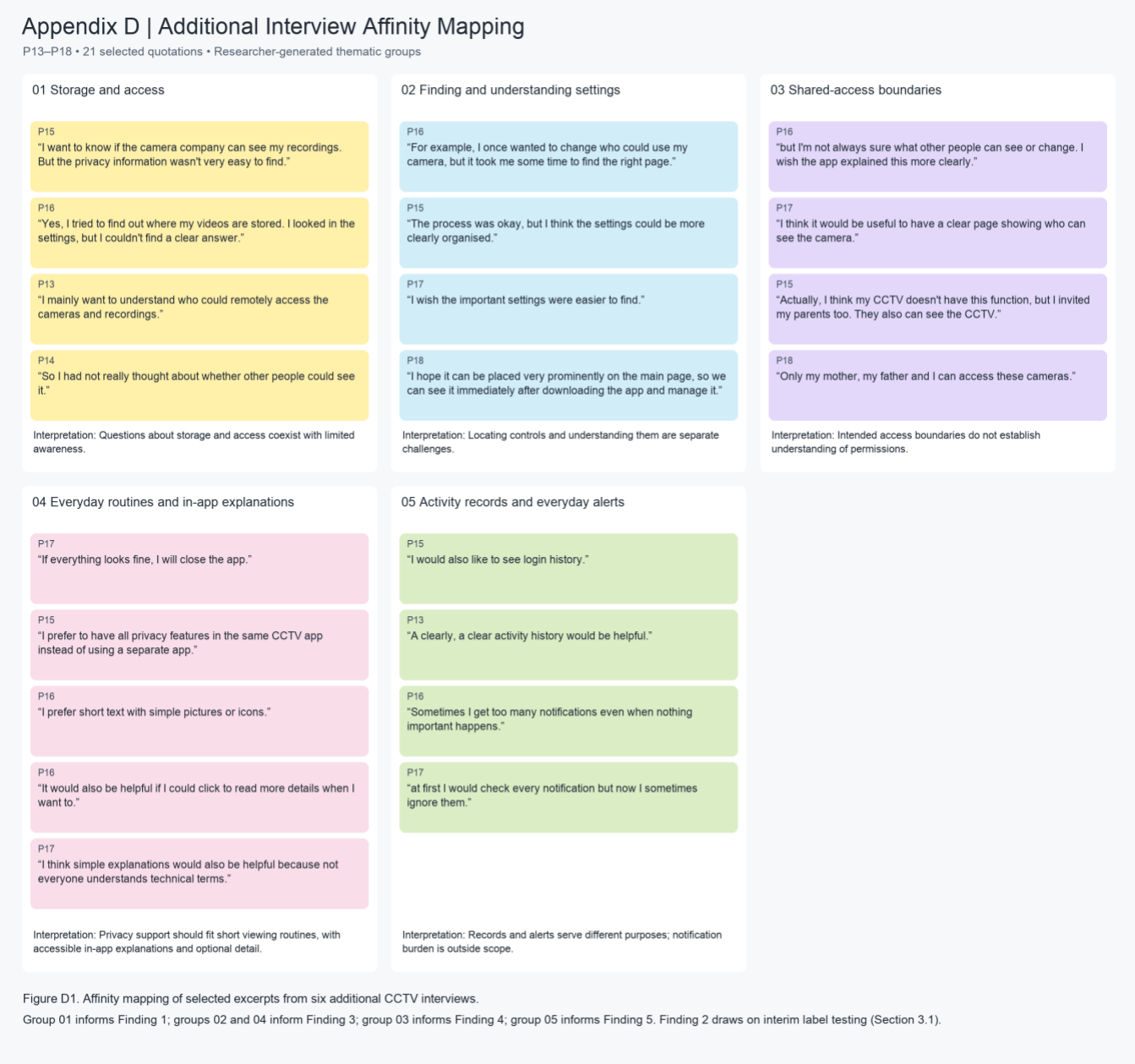


Figure D1. Affinity mapping of selected excerpts from six additional CCTV interviews.
Group 01 informs Finding 1; groups 02 and 04 inform Finding 3; group 03 informs Finding 4; group 05 informs Finding 5. Finding 2 draws on interim label testing (Section 3.1).

Figure D1. Affinity mapping of selected excerpts from six additional CCTV interviews. P13 describes store use; the other participants describe household use. Group 01 informs Finding 1; groups 02 and 04 inform Finding 3; group 03 informs Finding 4; group 05 informs Finding 5. Finding 2 draws on interim label testing.

Appendix E - Evidence to Design and Evaluation Index

This index distinguishes interview findings, team design responses and untested outcomes. Functional controls in the eufy concept require manufacturer cooperation for real implementation.

Table E1. Evidence, design responses and evaluation needs.

Finding and evidence	Requirement	Prototype response	Evaluation	Remaining limitation
F1: R1-P02 conditional-purpose question; P15 manufacturer-access question	R1	Data-flow explanations; Agreement; Storage & Backup	Explain storage, transmission and access conditions	Official sources may be incomplete; no verified technical protection.
F2: interim label interpretation observations	R5	Explanations of score meaning, factors and uncertainty	Explain rating direction and evidence limits before help	Scores are illustrative; summary measures require reconciliation.
F3: P15 in-app preference; P16 information finding and optional detail	R2, R5	Question-based guidance and links; layered explanations	Find settings; compare AIRIS with an in-app route	Separate-platform value unproven; tutorials planned.
F4: P16 uncertainty; P17 access-page request; multi-user literature	R3	Viewing Permissions; clip-specific Camera Events requests	Judge effective access and explain a change	Exact masking options are design hypotheses.
F5: P3 footage-access question; P15 login-history request	R4	Illustrative footage-viewing history; login history not implemented	Distinguish recording, viewing and login events	Evidence does not establish comprehensive logging needs.

E2 Participant support for Figure 3.2

Each record is counted once per theme; a record may support several themes. P13 is the store participant. The notification theme concerns unwanted or unexplained motion alerts, not an unexplained device indicator. P18's rare indicator event is therefore not counted in that theme.

Table E2. Participant-level support and source locations for the follow-up theme chart.

Theme	Supporting records	Count	Source and interpretation
In-app privacy entry	P13, P15, P16, P17, P18	5/6	P13.txt 07:40–07:56; P15 Q13; P16 Q13; P17 Q13; P18 privacy-entry answer. P18 requested a main-page entry rather than explicitly rejecting a separate app.
Unwanted or unexplained motion alerts	P14, P15, P16, P17	4/6	P14.txt: unexpected movement alerts; P15 Q10; P16 Q5/Q11; P17 Q10/Q11. All four are household records.
Privacy information or settings hard to find	P15, P16, P17	3/6	P15 Q7; P16 Q5/Q7; P17 Q5/Q6.
Clearer shared-access visibility or control	P13, P16, P17	3/6	P13.txt 04:37; P16 Q9; P17 Q9. P13 is contextual store evidence.
Text with pictures or icons preferred	P16, P18	2/6	P16 Q14; P18 presentation-preference answer. This is stated preference, not demonstrated comprehension.
Login or activity history requested	P13, P15	2/6	P13.txt 03:13 and 05:32; P15 Q10. Different requested event types are not treated as the same feature.

Appendix F - AIRIS Interface Screens

Screenshots supplied by the team on 27 October 2026 document the displayed prototype. They show interface content and proposed controls, not evidence of comprehension, technical enforcement or a commercial eufy partnership. The AIRIS model-guide screenshot appears in Figure 3.4a.

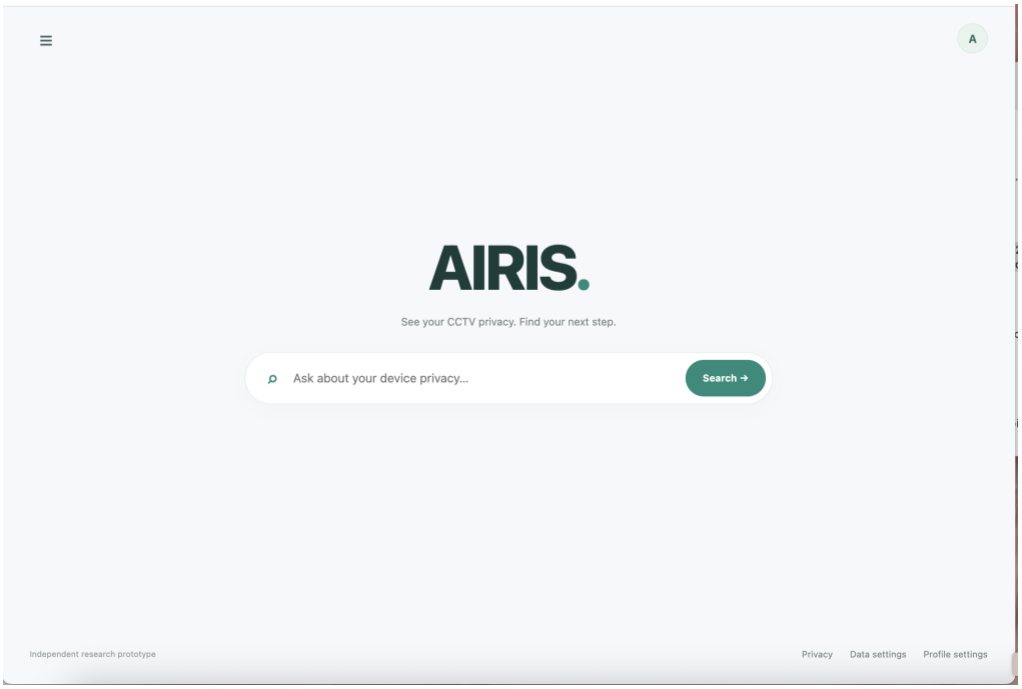


Figure F1. AIRIS home page prioritises a privacy-question search entry and identifies the site as an independent research prototype.

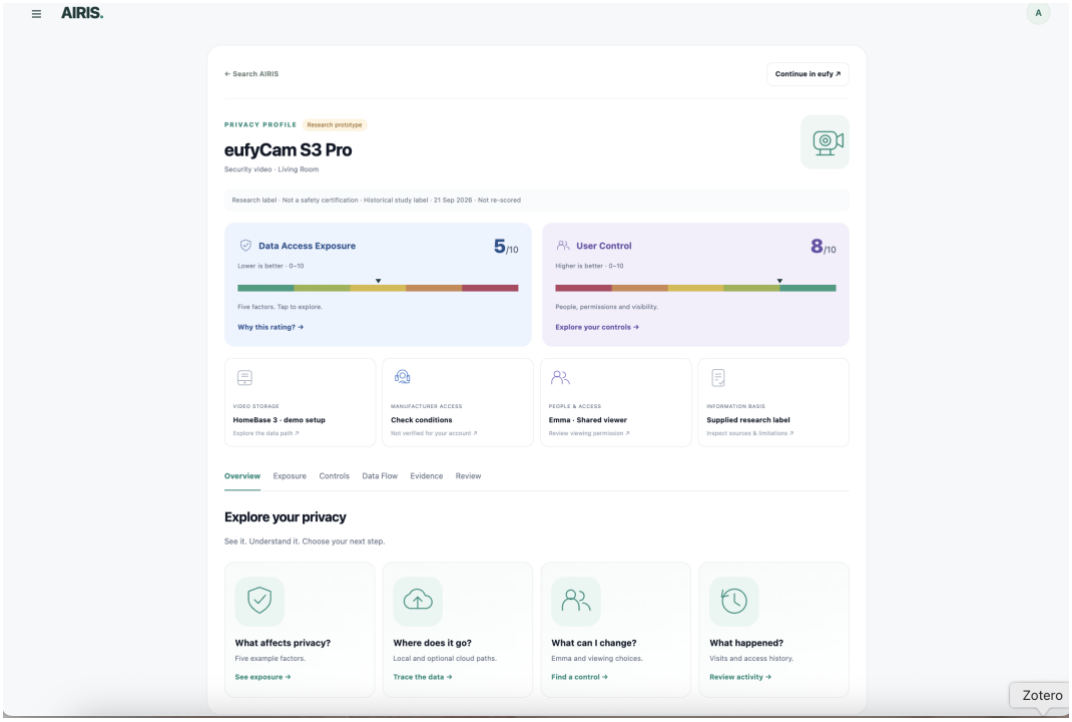


Figure F2. AIRIS Privacy Profile separates score directions, displays the research-label limitation and links to storage, access, evidence and review. The scores are historical study values, not current certified product ratings.

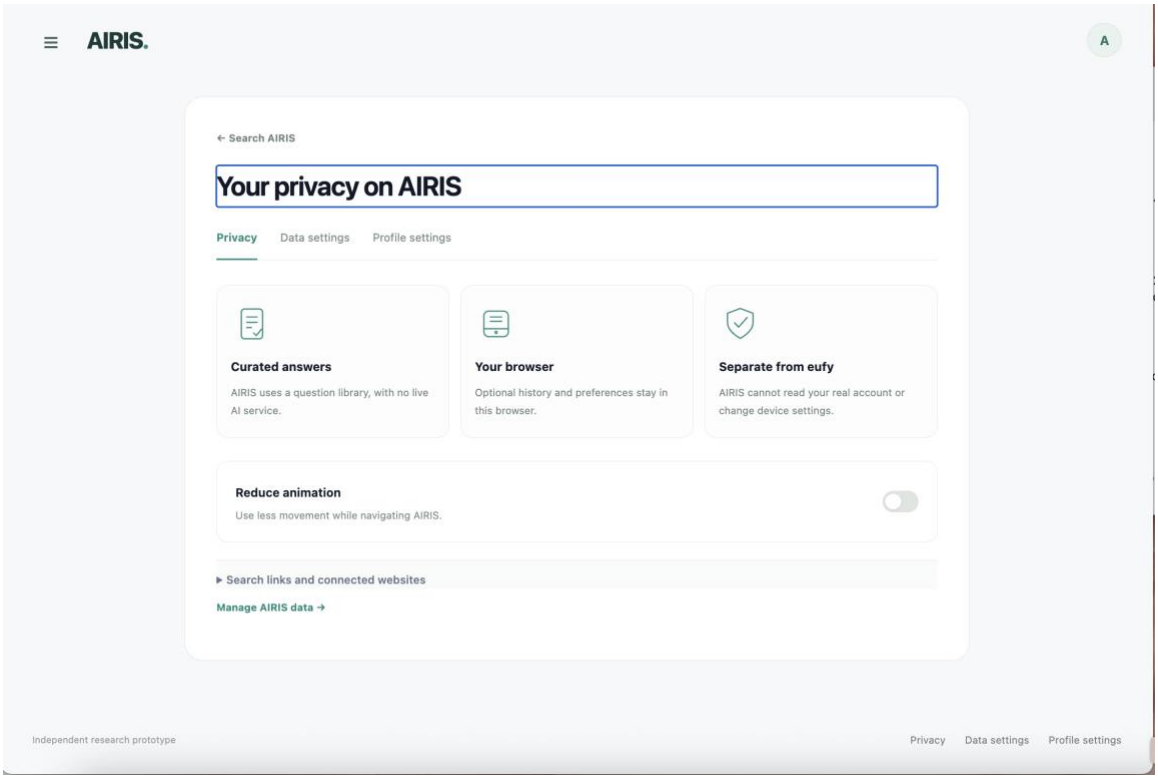


Figure F3. AIRIS Privacy screen explains curated answers, browser-held preferences and separation from eufy. A reduced-animation option is displayed.

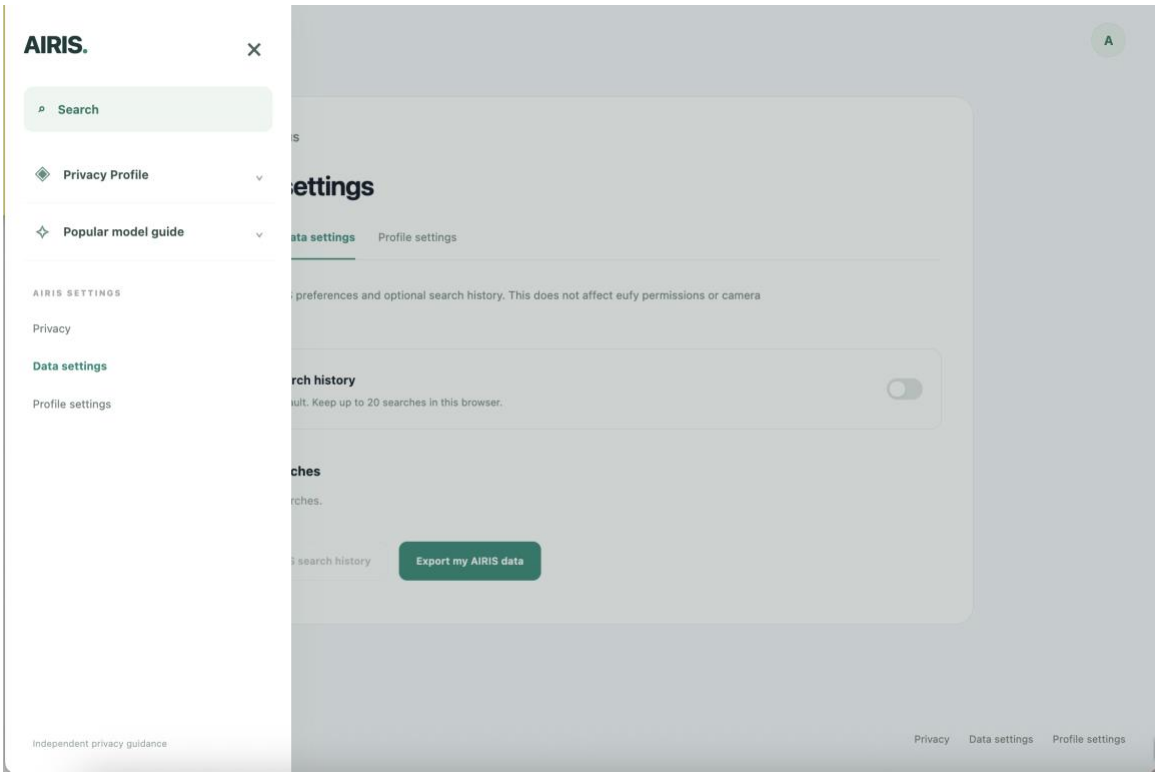


Figure F4. AIRIS navigation separates search, privacy profiles and model guides from its own settings. The underlying Data settings page shows optional search-history and data-export controls; these do not change eufy permissions.

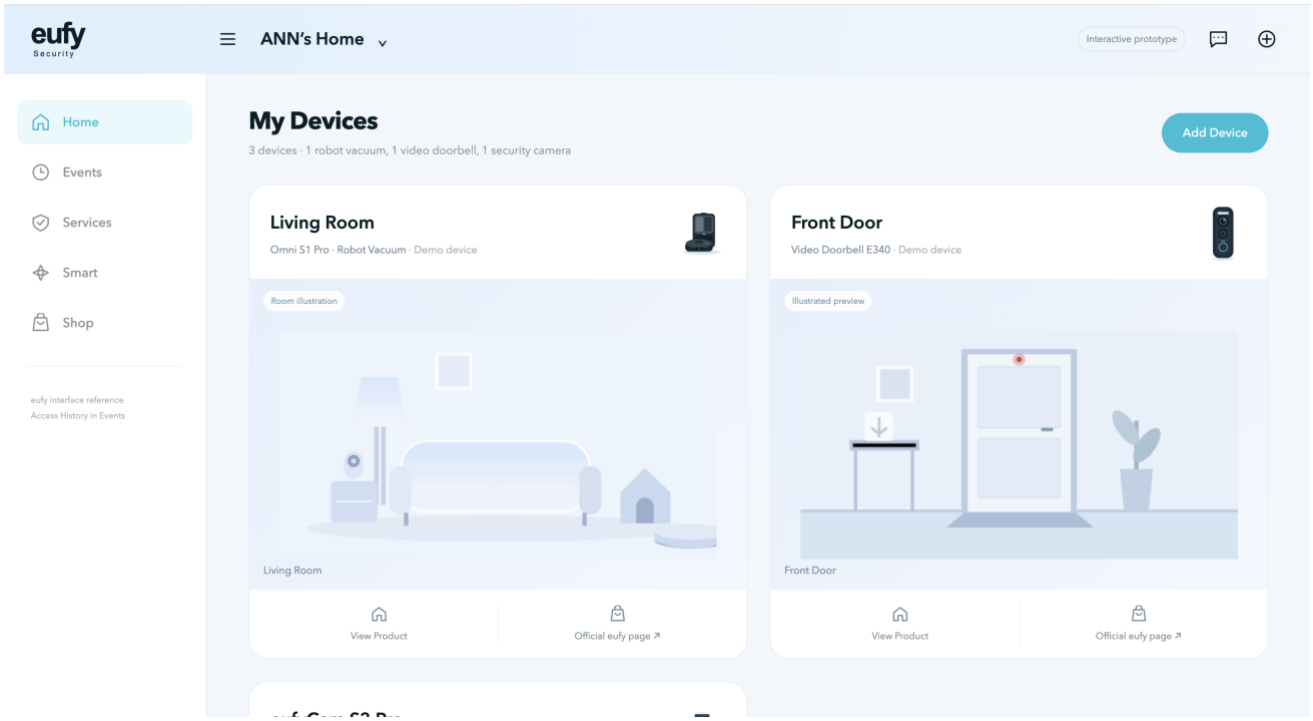


Figure F5. Latest eufy-facing Home lists demonstration devices. The report evaluates the home CCTV scenario; the robot vacuum is outside that evaluation scope. This is a proposed redesign, not an official eufy screen.

Appendix G - Current eufy Prototype Screens

Latest screenshots supplied by the team on 27 October 2026 replace earlier web-interface captures. They document the displayed design, including disabled and enabled agreement states, not completed user evaluation or real device enforcement. Historical paper-label stimuli in Appendix B remain separate evidence of earlier testing.

Figure G1. Privacy Agreement overview is reproduced in Figure 3.4c; the duplicate screenshot is omitted here.

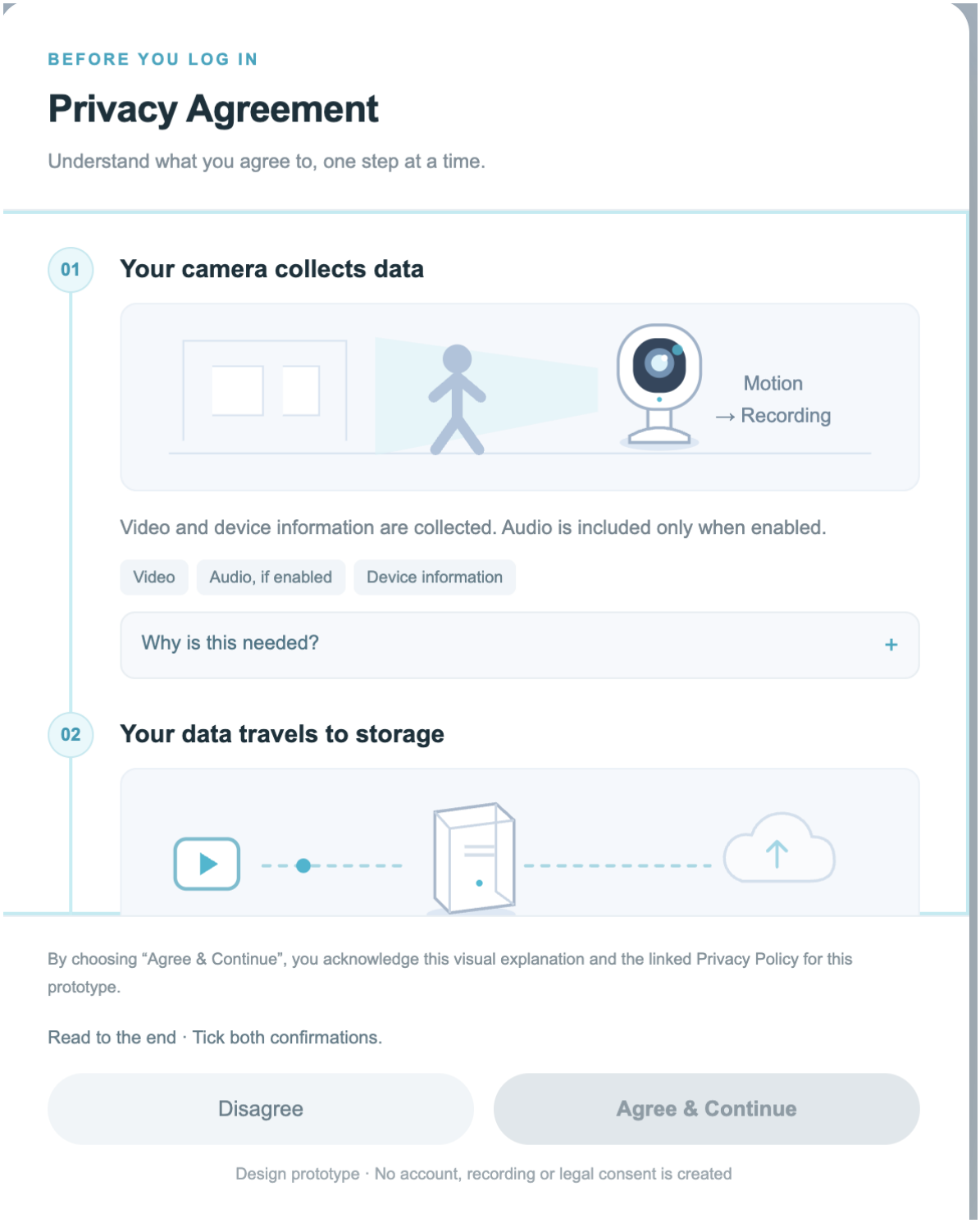


Figure G2. Data collection explanation. Video and device information are listed; audio is conditional on being enabled. The “Why is this needed?” explanation is expandable.

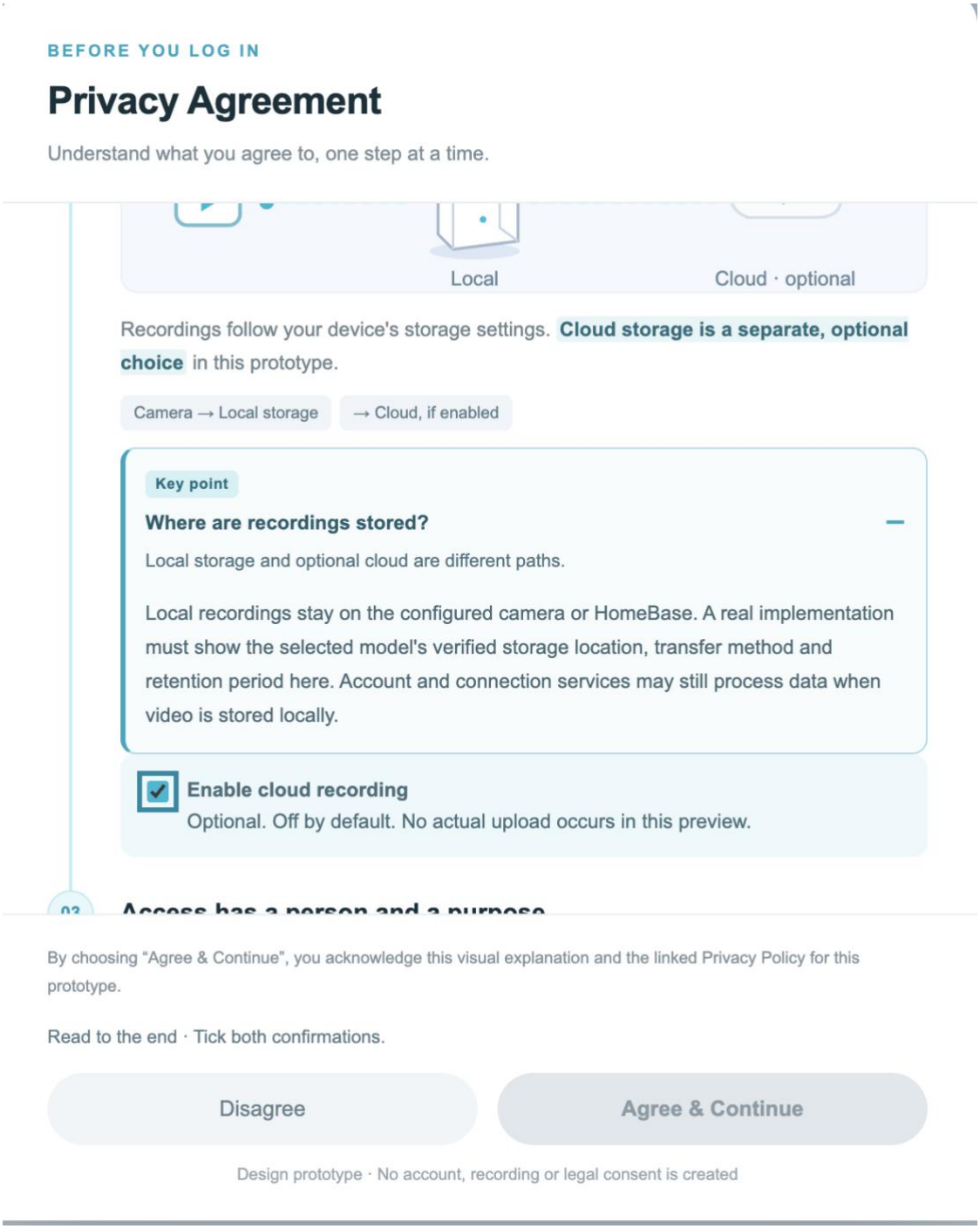


Figure G3. Storage explanation with the optional cloud choice selected. The interface states that no upload occurs in the preview and that local storage does not eliminate account or connection processing.

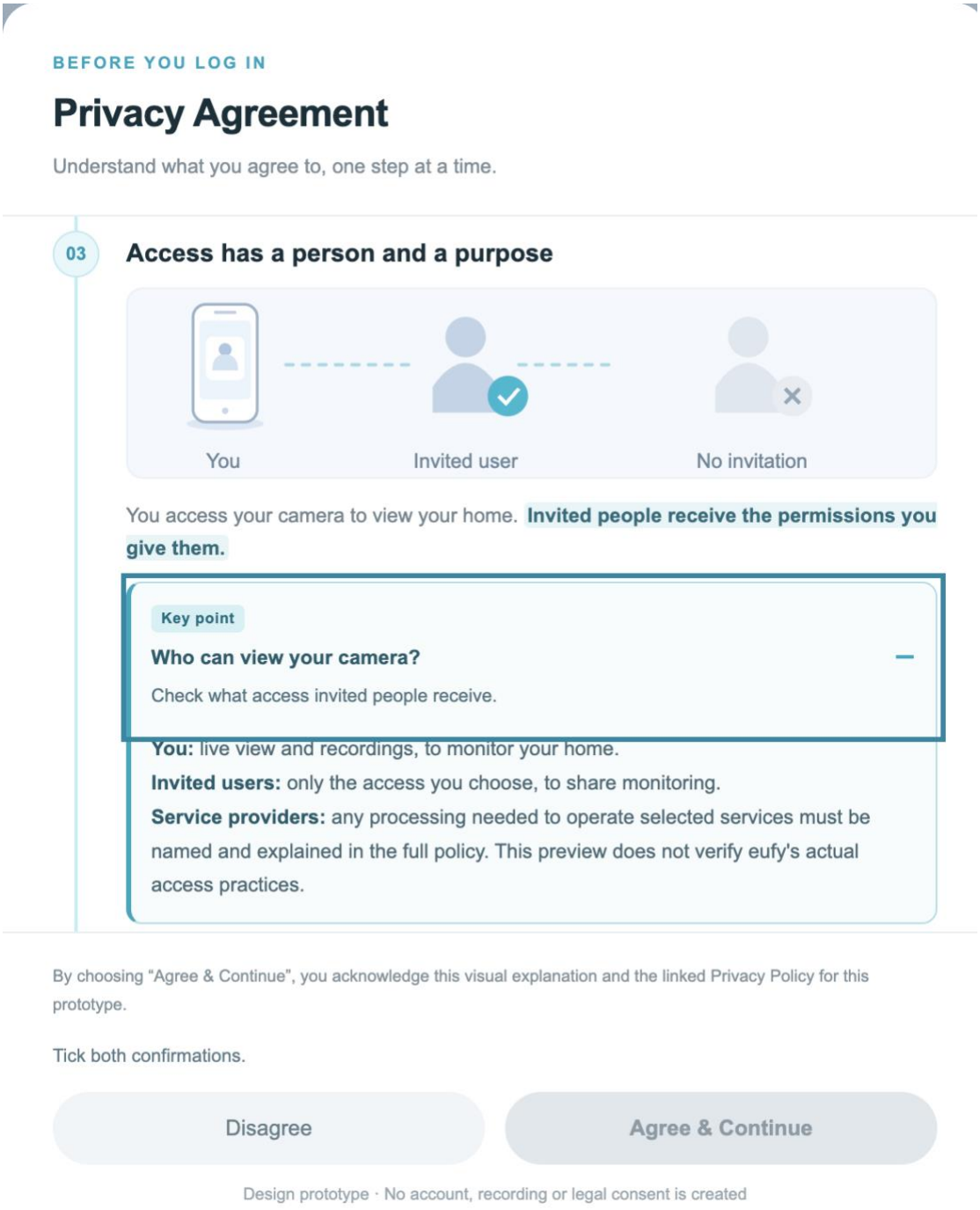


Figure G4. Access explanation distinguishes the owner, invited users and service providers. The preview explicitly does not verify eufy's actual access practices.

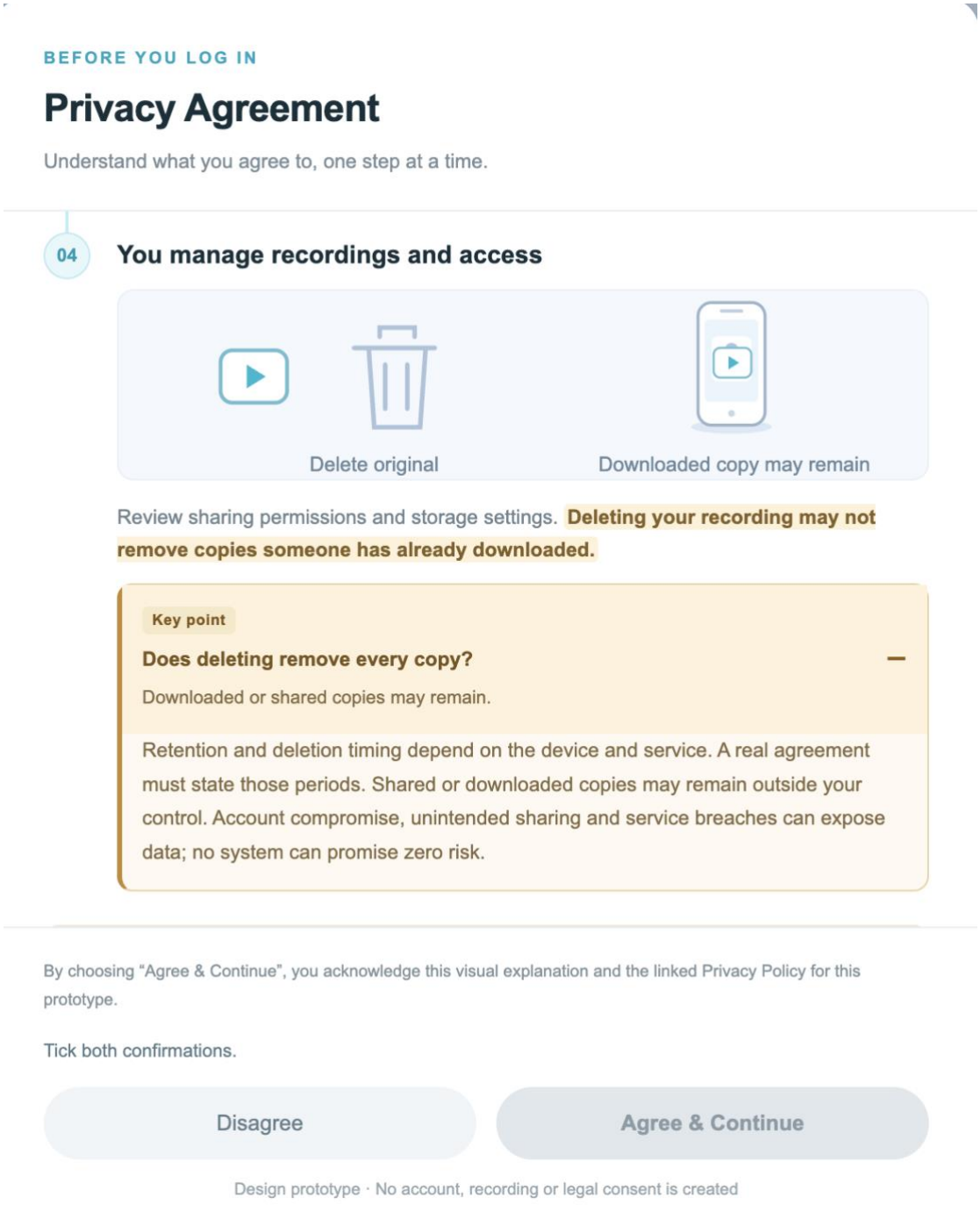


Figure G5. Deletion explanation states that downloaded or shared copies may remain. Retention and deletion periods depend on the device and service.

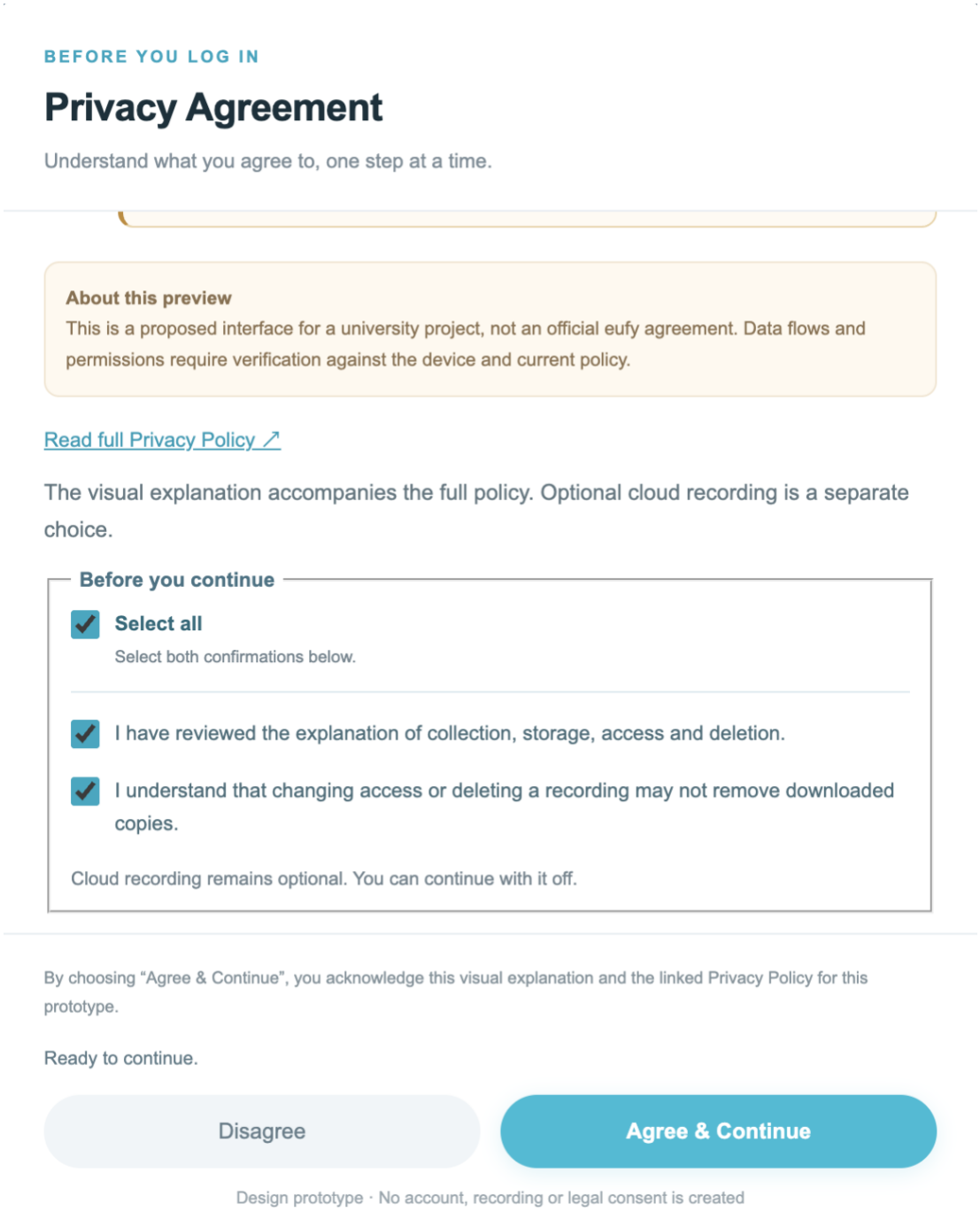


Figure G6. Confirmation state shows the full-policy link, two selected confirmations, optional cloud wording and an enabled continuation action. This state demonstrates interaction, not proven comprehension or legal consent.

Figure G7. Home device list is reproduced in Figure F5; the duplicate screenshot is omitted here.

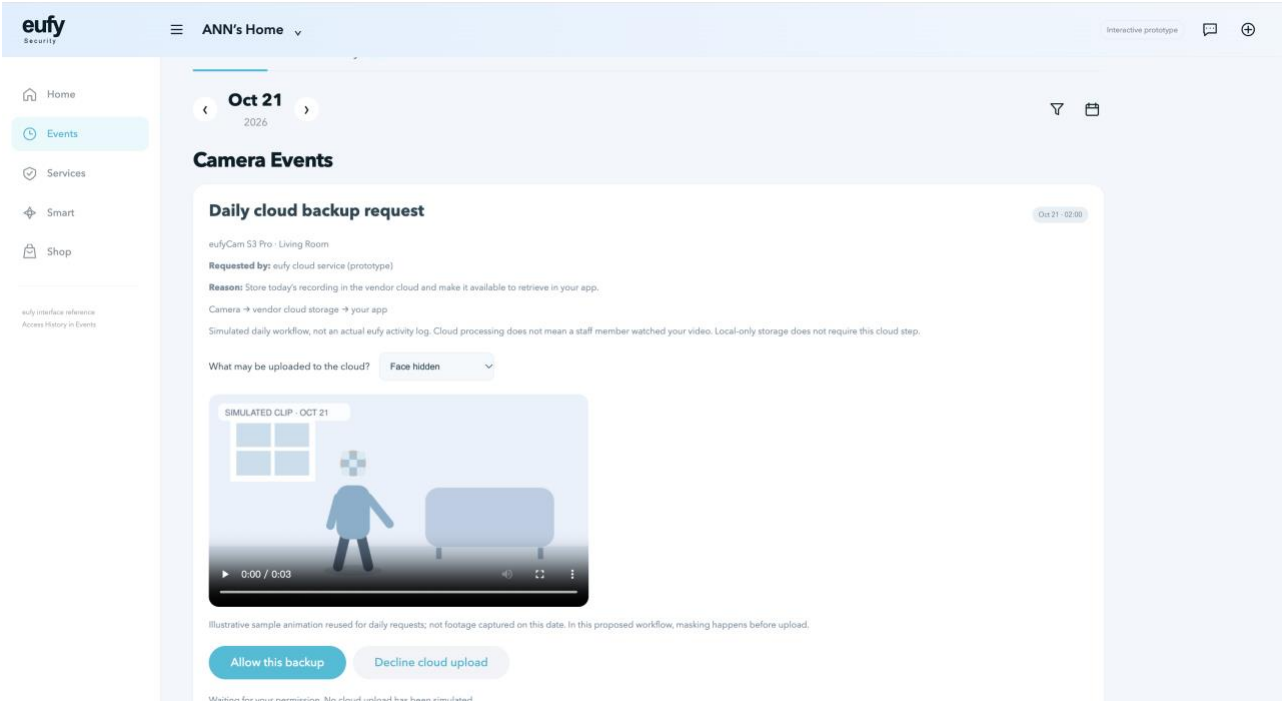


Figure G8. Camera Events presents a simulated cloud-backup request, requester, reason, data route, masking choice and allow/decline actions. Cloud processing is explicitly distinguished from a staff member watching footage.

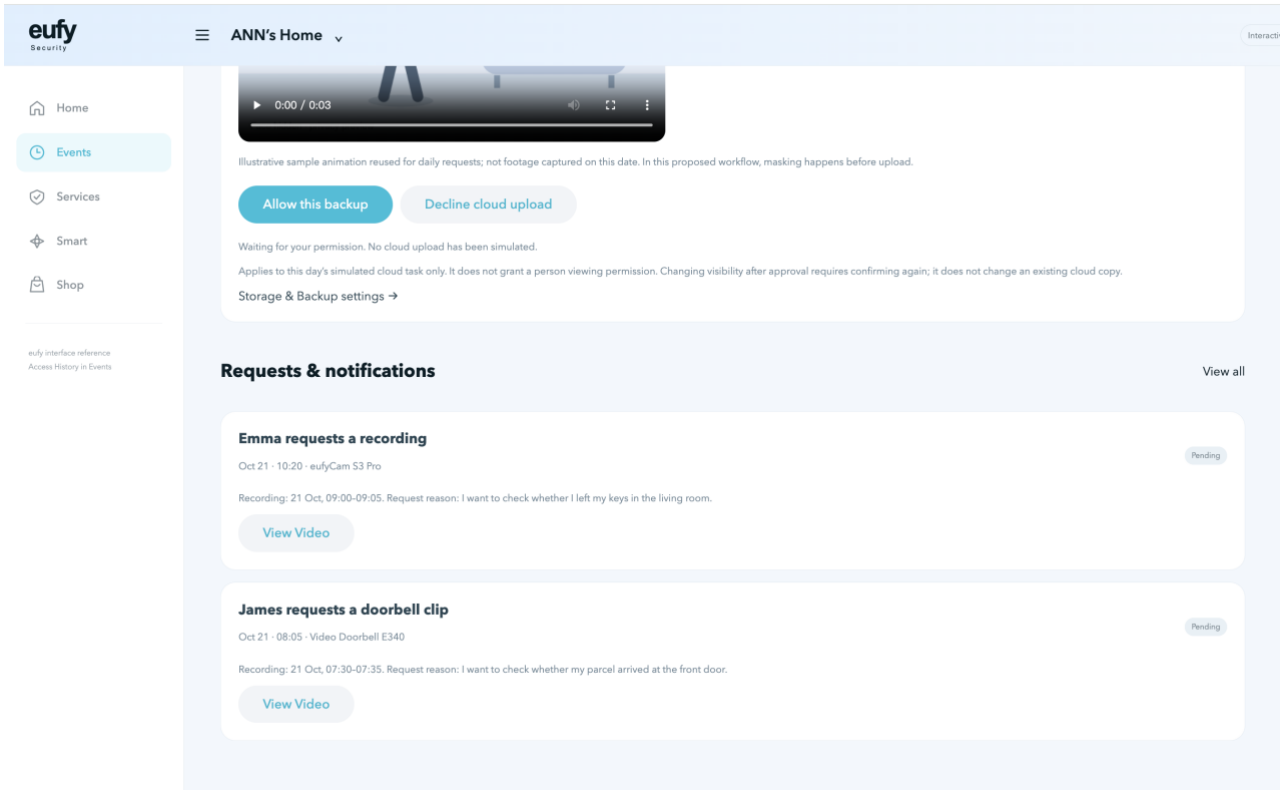


Figure G9. Requests & notifications distinguishes pending personal clip requests from the cloud-backup decision. Approval of one simulated cloud task does not grant a person viewing permission.

Figure G10. Access History is reproduced in Figure 3.4b; the duplicate screenshot is omitted here.

Appendix H – Product Privacy Labels

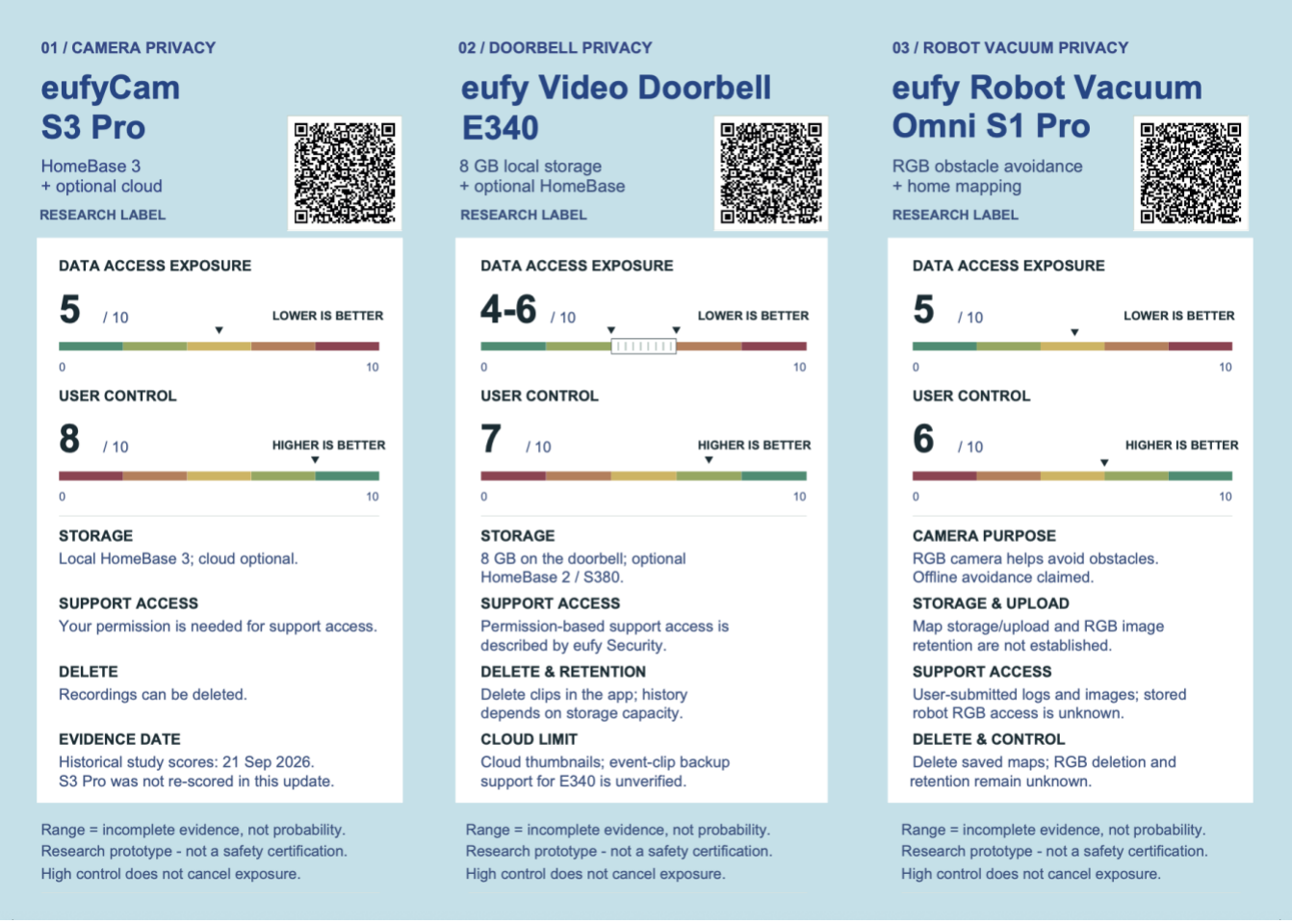


Figure H1. Privacy labels presented in the AIRIS exhibition brochure, summarising Data Access Exposure, User Control and product-specific information. These are research-prototype labels, not safety certifications, and have not undergone user testing.

Appendix I - AI Acknowledgement Cover Sheet

Your details

Binyan Zhang 48878162

Xinyue Chang 49372528

Yining Zhu 49053881

Yuxuan Yang 48950255

Assessment details

Course: DECO7380 Design Computing Studio 3
Assignment: Assessment 1.3 Exhibit and Final Project Proposal
Team: ANN

AI use

[X] Yes, AI tools have been used to generate or edit material in this document.
[ ] No, AI tools have not been used to generate or edit material in this document.

Details of use

The representative prompts below are English renderings of requests made in this report-support conversation. Dates refer to documented assistance on 27 October 2026. The table is an editable account of observed use, not a declaration that the team has already completed every source check.

Team member	Tool	Use	Prompt(s)	Section	Date
Yining Zhu	ChatGPT / Codex	Generated and revised responsive prototype code using the supplied Figma prototype and eufy screenshots. Updated layouts, product cards and animations following my feedback.	Implement my History prototype as a website that works on desktop and mobile. Keep the eufy visual style and reuse the scene illustrations for the three Home products.	Prototype /history	21–26 Oct 2026
Yining Zhu	ChatGPT / Codex	Implemented the privacy information pages, expandable explanations and links to official policies.	Organise the privacy content into Your data journey, People & access, and Keep, share or delete. Also retain general privacy information and policies.	Prototype	21–22 Oct 2026
Yining Zhu	ChatGPT / Codex	Generated and revised Access History, viewing-permission controls and a simulated viewer face-verification interaction.	Show who viewed the cameras and whether access is allowed. Add face verification for the person requesting access, rather than people appearing in recordings.	Prototype /history	21–26 Oct 2026

Team member	Tool	Use	Prompt(s)	Section	Date
Yining Zhu	ChatGPT / Codex	Modified the Agreement interaction, including reading requirements, confirmation checkboxes, Select all and highlighted expandable content.	Enable Agree & Continue only after scrolling to the bottom, waiting ten seconds and selecting both confirmations. Add Select all and highlight important privacy explanations.	prototype/ agreement	26 Oct 2026
Yining Zhu	ChatGPT	Reviewed and refined an outline drafted by me, suggesting improvements to its structure, headings and section coverage.	“Review my report outline and suggest improvements to its structure, headings and coverage against the assessment brief.”	Report outline and section headings	26 Oct 2026
Yining Zhu	DeepL	Translated my Chinese notes and draft ideas into English	Translate my Chinese design notes into clear English for a university assignment.	Report	27 Oct 2026
Yining Zhu	Claude	Reviewed drafts for logic and evidence-chain gaps	Check this version for logic problems and gaps in the evidence chain.	Whole report	27 Oct 2026
Yining Zhu	ChatGPT	Checked grammar and improved the clarity of my English writing	Check report for grammar and spelling mistakes.	Report	27 Oct 2026
Xinyue Chang	ChatGPT	Helped modify small sections of code during prototype development.	How can I change this code to make the interaction work?	Prototype development	22 Oct 2026
Xinyue Chang	ChatGPT	Helped locate bugs and suggested possible fixes.	Can you help me find why this feature is not working?	Prototype debugging	23 Oct 2026
Xinyue Chang	ChatGPT	Explained unfamiliar code to help me understand it.	Can you explain what this part of the code does?	Prototype development	23 Oct 2026
Xinyue Chang	ChatGPT	Helped identify incomplete or unclear parts of the prototype.	Which parts of this prototype need further development or explanation?	Prototype review	23 Oct 2026
Xinyue Chang	ChatGPT	Corrected English grammar and improved sentence clarity.	Please check the grammar and make the wording clearer without changing my meaning.	Report and prototype interface text	25 Oct 2026
Yuxuan Yang	ChatGPT / Codex	Helped organise the privacy agreement into illustrated sections covering data collection, storage, access and deletion.	Divide the privacy agreement into short sections, with a clear explanation and illustration for each topic.	Prototype — Privacy Agreement visuals	20 Oct 2026
Yuxuan Yang	ChatGPT / Codex	Helped create SVG/CSS illustrations showing the camera, storage, cloud and phone, alongside the written explanations.	Create simple diagrams showing how camera data moves between devices and storage services.	Prototype — Privacy Agreement visuals	20 Oct 2026

Team member	Tool	Use	Prompt(s)	Section	Date
Yuxuan Yang	ChatGPT / Codex	Helped implement expandable explanations, confirmation checkboxes and the Agree / Disagree interactions.	Add optional details and clear confirmation states, and update the Continue button when the required choices are completed.	Prototype — Privacy Agreement interactions	20 Oct 2026
Yuxuan Yang	ChatGPT	Helped prepare a before-and-after prototype comparison figure to illustrate design changes.	Create a comparison figure showing the earlier and current prototype designs.	Section 3.3	27 Oct 2026
Yuxuan Yang	ChatGPT	Helped explain alternative designs and trade-offs, including separate AIRIS guidance versus guidance within the CCTV app.	Explain why the current interaction was chosen and discuss its alternatives and limitations.	Sections 3.3–3.4	27 Oct 2026
Binyan Zhang	ChatGPT / Codex	Helped modify small sections of code for prototype interactions.	How can I adapt this code to support the CCTV privacy settings interaction?	Prototype development	20/10/2026
Binyan Zhang	ChatGPT / Codex	Helped identify bugs and resolve navigation issues.	Why does this button fail to open the intended prototype screen, and how can I fix it?	Prototype debugging and navigation	22/10/2026
Binyan Zhang	ChatGPT	Explained unfamiliar code and how it works.	Can you explain how this code handles user input and updates the privacy settings?	Prototype development	23/10/2026
Binyan Zhang	ChatGPT	Helped identify unclear or incomplete parts of the prototype.	Which parts of this CCTV privacy prototype need clearer feedback or further development?	Prototype review	23/10/2026
Binyan Zhang	ChatGPT	Helped improve English grammar, wording and clarity.	Please edit this report paragraph for grammar and clarity while preserving its meaning.	Report editing	25/10/2026

Review before submission: verify interview wording and participant IDs, check cited sources, reconcile the test-record discrepancies identified in Appendix B4, and confirm the report's implementation claims against the team's prototype. AI-generated explanations and suggested citations are not research data.

Appendix J - Product Information Sources

These sources support how the report describes claims and conditions; they do not verify device behaviour or validate prototype features. The Indoor Cam E30 entry is a model-specific example, not evidence about the eufyCam S3 Pro shown in the current prototype.

Source	Relevant statement or reporting	Report use and limit
eufy (n.d.-b), Privacy commitment	Describes local recording and optional cloud services. Thumbnail notifications can involve cloud processing. Default transit encryption and optional Advanced Encryption are distinguished.	Sections 1.2 and 4.2: explain conditions. These are manufacturer statements, not our security findings.

eufy (n.d.-a), Indoor Cam E30	Lists microSD or HomeBase 3 local storage and optional cloud storage; its footnote describes cloud use for selected features.	Illustrates model-specific conditions. Do not transfer these storage options or limits to other models.
Lovejoy (2022), 9to5Mac	Reports the cloud-thumbnail controversy and Anker's response, including its denial that user data had been exposed.	Historical motivation in Section 1.2. It is not evidence of a current vulnerability or every product's behaviour.
Lovejoy (2023), 9to5Mac	Reports Anker's admission concerning unencrypted web live streams and promised encryption updates, audits and a bug bounty.	Historical context. App streams and web recordings are distinguished from the reported web live-stream issue.

Full page titles and URLs appear in References. For deployment, each AIRIS answer would need a model-specific source, configuration conditions, a review date and an explicit statement of what has not been verified. The current research label is historical and has not been re-scored from these pages.

Appendix K – Prototype Repository

The GitHub repository contains the source code for the AIRIS and conceptual eufy web prototype. The prototype demonstrates proposed interactions using sample data and does not control real CCTV devices or accounts.

Repository: https://github.com/yenyong1211/DECO7380_CCTV

Appendix L – Exhibition Poster

